# All in One: A Unified Network for Hyperspectral Image Fusion

Yaoting Liu, Xinwei Cheng, Hui Li, Shenglian Luo, and Bin Yang, *Member, IEEE*

***Abstract*— Hyperspectral image fusion (HIF) is pivotal in scientific research and engineering. With the fast development of deep learning techniques, numerous HIF algorithms have been proposed. However, traditional HIF algorithms, which are designed specifically for different input conditions, such as hyperspectral–multispectral image fusion (HMF), hyperspectral-panchromatic image fusion (HPF), and hyperspectral–multispectral-panchromatic image fusion (HMPF), cause significant inconvenience in the practical application of HIF. To overcome this issue, we propose a unified network (UniNet) that integrates HMF, HPF, and HMPF in an "all in one" manner. During testing, UniNet can perform HIF under different input conditions, enhancing its practicality. Specifically, the UniNet consists of the HMF, the HPF, and the HMPF sub-networks. Each sub-network is designed to reconstruct a high-resolution (HR) hyperspectral image (HSI) under a specific input condition. A mutual learning strategy is proposed to establish the relationships among sub-networks, enabling the transfer of spectral–spatial knowledge across different input conditions. In addition, a multiscale gradient spatial attention module (MGSAM) is designed in the network to better extract the spatial edge details of HSIs at multiple scales. Once trained, UniNet is capable of working with its sub-networks independently. Extensive experiments show that UniNet outperforms 18 state-of-the-art algorithms with five HSI datasets. Furthermore, the effectiveness of our algorithm is validated through ablation experiments.**

***Index Terms*— Hyperspectral image fusion (HIF), multiscale (MS) gradient spatial attention (GSA), mutual learning, unified network (UniNet).**

## I. Introduction

HYPERSPECTRAL image (HSI) records the radiation reflected from objects with abundant spectral information from visible to near-infrared wavelength [1], [2]. It is extremely significant for the development of remote sensing techniques due to the fine spectral details, which play a crucial role in tasks such as anomaly detection [3], [4], change detection [5], [6], land cover classification [7], [8], and so on. Nevertheless, limited by the signal-to-noise constraints of optical sensors, higher spectral resolutions come at the expense of lower spatial resolution, consequently making it inappropriate for applications at finer scales [9]. Therefore, the fusion of high-resolution (HR) panchromatic image (PAN) or mid-resolution (MR) multispectral image (MSI) and low-resolution (LR) HSI has gained significant attention, owing to its potential to obtain the high spatial resolution HSI [10]. The integration of these diverse images presents a remarkable opportunity to capture both spatial and spectral details simultaneously, providing valuable insights for a wide range of applications of HSIs.

Benefiting from the rapid development of deep learning in the field of image fusion, hyperspectral image fusion (HIF) also achieved excellent success. Current deep learning-based HIF algorithms focus on hyperspectral–multispectral image fusion (HMF), i.e., fusion via HSI and MSI to obtain the MR HSI. Specifically, Zheng et al. [11] proposed an edge-conditioned feature transform network for HMF, which obtains better performance by incorporating the spatial edge prior of MSIs. To capture the global relationship of the images, Hu et al. [12] proposed a transformer-based fusion network (Fusformer) that captures the relationship between long-range pixels via self-attention. However, the spatial resolution of MSIs is still relatively low. To obtain higher spatial resolution HSI, researchers have explored the fusion of higher resolution images such as PAN, i.e., fusion via HSI and PAN to obtain the HR HSI. He et al. [13] proposed a spectrally predictive convolutional neural network (CNN) called HyperPNN to fuse HSI and PAN, which strengthens the spectral predictive capability of hyperspectral pansharpening. To enhance the representation ability of the network, a multistage dual-attention guided fusion network (MDANet) [14] is designed and attains favorable fusion performance in hyperspectral-panchromatic image fusion (HPF). Nevertheless, HPF brings problems such as spectral distortion due to the significant spectral difference between PAN and HSI. Therefore, fusion via HSI, MSI, and PAN to obtain HR HSI has received considerable attention in recent years, which can exploit the advantages of MSIs and PANs simultaneously to obtain satisfactory HR HSIs [15], [16].

Nevertheless, perfection remains elusive. The reconstruction of HSIs with high spatial and spectral resolution simultaneously remains a formidable task due to the following reasons.

Manuscript received 17 January 2024; revised 8 May 2024; accepted 6 June 2024. Date of publication 14 June 2024; date of current version 27 June 2024. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFD1700802; in part by the Key Research and Development Program of Hunan Province, China, under Grant 2023SK2041; in part by the Natural Science Foundation of Hunan Province, China, under Grant 2023JJ20018; and in part by Changsha Science and Technology Program under Grant kq2208036. *(Corresponding author: Bin Yang.)*

Yaoting Liu, Xinwei Cheng, and Bin Yang are with the College of Electrical and Information Engineering and the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Hunan University, Changsha 410082, China (e-mail: liuyt1998@hnu.edu.cn; chengxinwei@hnu.edu.cn; binyang@hnu.edu.cn).

Hui Li is with the School of Environment, Beijing Normal University, Beijing 100875, China (e-mail: lihuui@bnu.edu.cn).

Shenglian Luo is with the Key Laboratory of Jiangxi Province for Persistent Pollutants Control and Resources Recycle, Nanchang Hangkong University, Nanchang 330063, China (e-mail: sllou@hnu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3414438

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

1) Traditional HIF algorithms typically require training separate models for different input conditions. Therefore, it is necessary to design an algorithm that can handle multiple input conditions while producing high-quality HSI.
2) The networks that generate fused HR HSI in different input conditions are usually optimized independently. Therefore, an appropriate algorithm is required to facilitate effective knowledge transfer and optimization among different sub-networks.
3) The correlation between multiscale (MS) feature representation and spatial edge details is usually ignored in deep learning-based algorithms for HIF. To obtain high-quality HR HSI, the edge details of the HSI at different scales are essential.

To address these drawbacks, in this article, we propose a unified network for HIF, called UniNet, to overcome the problems under different input conditions. This is an "all in one" network, i.e., a UniNet that includes HMF, HPF, and HMPF simultaneously. The UniNet consists of three sub-networks, the first sub-network aims to fuse HSI and MSI, the second sub-network aims to fuse HSI and PAN, and the third fusion network can fuse all MSI, PAN, and HSI. The difference between the UniNet and the existing HMF, HPF, and HMPF algorithms is shown in Fig. 1. The UniNet acts as the intersection of the existing algorithms. It can tackle different input conditions for HIF, facilitating the application of HIF via integrating multiple HIF tasks into a UniNet. To optimize the UniNet, a mutual learning strategy is proposed to enable the fused images generated in different input conditions to learn from each other. The mutual learning strategy allows knowledge transfer among different sub-networks via loss function constraints and does not require direct network connections. Moreover, an MS gradient spatial attention module (MGSAM) is designed to enable the network to preserve spatial edge details of HSIs at multiple scales. The main contributions of this work are given as follows.

1) We propose a UniNet that can tackle different input conditions for HIF in an "all in one" manner. To the best of our knowledge, this is the first attempt to integrate multiple HIF tasks into a UniNet.
2) A mutual learning strategy is proposed for HIF that helps to transfer knowledge and experience of spatial and spectral information among sub-networks, thus improving network optimization.
3) An MGSAM is designed to efficiently extract spatial edge details from the images in a fine-and-coarse fashion.

The rest of the article is divided into four sections. Section II introduces the related work on HMF, HPF, and HMPF. Section III describes the proposed UniNet in detail. Section IV presents the experiments and results. Section V is the conclusion.

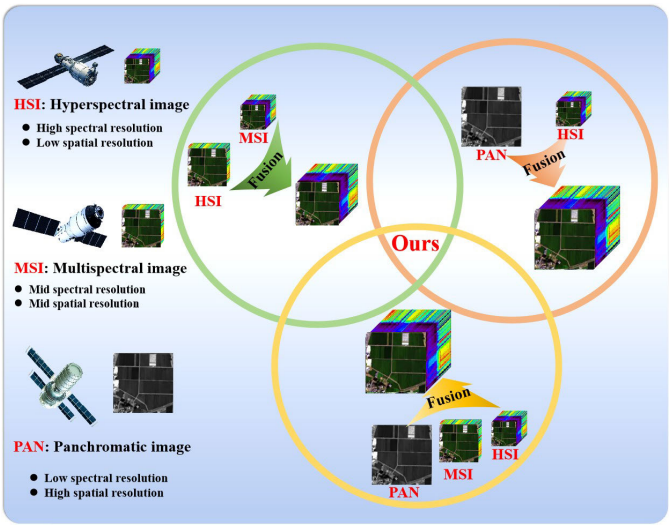



Fig. 1. Flowchart shows the difference between the proposed UniNet and the existing HMF, HPF, and HMPF algorithms. The proposed UniNet acts as the intersection of HMF, HPF, and HMPF and enables HIF for all three input conditions.

## II. Related Work

### A. HMF

The most popular HIF task is the fusion of MSIs and HSIs. The algorithms of HMF can be generally categorized into Bayesian-based algorithms [17], [18], tensor-based algorithms [19], [20], matrix-factorization-based algorithms [21], [22], and deep learning-based algorithms. The Bayesian algorithm uses the Bayesian theory to establish a probabilistic model that achieves the fusion of HSIs with MSIs by estimating the posterior probability of the fusion result. Based on this theory, Eismann and Hardie [17] proposed the maximum a posteriori estimation algorithm for HMF. Akhtar et al. [18] addressed the problem of HIF by incorporating sparse representation. These algorithms have certain requirements for prior knowledge, but prior knowledge often relies on assumptions; thus, they usually cannot flexibly adapt to different HSIs. The tensor-based algorithms utilize tensor decomposition techniques to represent HSIs and MSIs as low-rank tensors and then reconstruct the fusion results. To learn the correlation between spatial, spectral, and nonlocal modes of nonlocal similar HR HSIs, a low tensor-train rank (LTTR)-based HIF algorithm is proposed by Dian et al. [19]. He et al. [20] proposed a novel algorithm to simultaneously learn the high spectral resolution core tensor of HSIs and the high spatial resolution core tensor of MSIs and obtained HR HSI by tensor ring representation reconstruction. The matrix-factorization-based algorithms represent the HSI and MSI as two matrices, respectively, and the fusion is implemented by matrix decomposition and reconstruction. To mine the underlying manifold structure in the spatial domain of HR HSIs, Zhang et al. [21] proposed to regularize the HIF problem by exploiting the self-expression property of HSI patches within the same cluster. In [22], a truncated matrix decomposition algorithm is proposed for HMF, which exploits the superpixel-level low-rank properties of HSIs.

Nowadays, with the popularity of deep learning, outstanding works based on deep learning for HMF have increasingly emerged. Palsson et al. [23] applied a 3-D CNN for HMF while utilizing dimensionality reduction to reduce computational complexity. Yet, cross-modal information between HSIs

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

and MSIs is still difficult to fuse. In [24], a spatial–spectral reconstruction network (SSRNet) is designed to enable a more adequate interaction of spatial and spectral information. However, the receptive field of CNN is limited. Deng et al. [25] proposed a novel pyramid shuffle-and-reshuffle transformer (PSRT) for HMF, which utilizes a pyramid structure based on window self-attention to effectively describe the long-range dependencies. To deal with the rich spectral information of LR HSIs and the complex structural features of HR MSIs, Pan et al. [26] proposed a structure-color preserving network to obtain HR HSIs. Meanwhile, Vivone [27] published a survey of HMF to provide a comprehensive overview of the current state of this field. Although HMF algorithms are popular, they encounter challenges in improving the spatial resolution of fused images. This limitation stems from the inherent drawback of MSIs, characterized by a relatively low spatial resolution.

### B. HPF

In contrast to MSI, PAN can provide higher spatial resolution. Therefore, PAN can contribute richer spatial details to LR HSI in the HIF task. In previous years, traditional algorithms occupied the mainstream of HPF such as multiresolution analysis [28], component substitution [29], and so on. Nevertheless, the learning ability of these traditional models is weak, thus leading to poor fusion performance. Meanwhile, these algorithms usually require complex manual design, which reduces the generalization ability of the models. With deep learning demonstrating superior performance in remote sensing image processing, it is natural to utilize CNNs directly to learn the fusion process of LR HSIs and PANs. Previous methods require different models to be trained for HSIs with different bands, and a novel network is proposed, which can sharpen HSI in any number of bands using a single pretrained model via fine-tuning the parameters of a small module [30]. In [31], an iterative super-resolution algorithm is proposed based on a deep HSI denoiser into a model-guided deep convolutional network for the fusion of the LR HSIs and HR PANs. In [32], a texture and spectral feature fusion converter is proposed to better transfer texture details from PAN to LR HSI for obtaining higher quality fused images. To further the development of HPF, Vivone et al. [33] organized the 2022 WHISPERS Hyperspectral Pan-Sharpening Challenge. As for HPF algorithms, they leverage PAN that boasts higher spatial resolution. However, a significant drawback arises from the spectral distortion introduced due to the substantial difference in spectral resolution between HSI and PAN.

### C. HMPF

As mentioned above, there are shortcomings of HMF and HPF, respectively. Therefore, HMPF is receiving considerable attention due to its potential to simultaneously use MSI and PAN, thereby mitigating the drawbacks of HMF and HPF through the amalgamation of their respective benefits. Yokoya et al. proposed an algorithm based on coupled nonnegative matrix factorization (CNMF) for the fusion of HSI, MSI, and PAN first [34]. Arablouei et al. [35] utilized forward observation and linear mixture models to resolve the HMPF task as an optimal maximum likelihood estimation problem. In [36], a computational algorithm is proposed, which subtly exploits the low-rank property of HSIs and the gradient similarity between MSIs and PANs. Nevertheless, the challenge of developing handcrafted parameters and inaccurate prior restricts the effectiveness of fusion. To obtain better fusion results, a deep network (HyperNet) is proposed for HMPF in [37], where CNNs provide excellent learning capabilities in attaining better fusion performance in HMPF. Neural networks in deep learning usually belong to black-box models without distinct physical meanings [38]. To improve the interpretability of the network, Tian et al. [16] designed a model-driven deep network and achieved favorable results. Despite the remarkable performances, a notable caveat in HMPF algorithms is that they are strict with input conditions, i.e., HSI, MSI, and PAN should be available simultaneously, thereby limiting their applicability.

## III. Methodology

### A. Motivation and Overview

As mentioned above, there are disadvantages to all three HIF algorithms. First, the quality of fused HSI from the HMF algorithms remains to be improved due to the relatively low spatial resolution of MSIs. Second, the HPF algorithms use PANs, which have a higher spatial resolution but suffer from spectral distortion due to the large spectral difference between the HSI and PAN. Third, the HMPF algorithms can alleviate the above two problems remarkably fine; however, it does not always have HSI, MSI, and PAN as input conditions. To overcome the limitations of these three algorithms, we proposed UniNet, which allows HMF, HPF, and HMPF, to be implemented separately in the testing phase under different input conditions. The proposed network can facilitate HIF by exploiting MSI and PAN as input conditions simultaneously and also subtly eliminates the disadvantage that HMPF cannot be implemented when MSI or PAN is unavailable.

Unlike previous HMF, HPF, and HMPF algorithms that require training separate models for different input conditions [39], UniNet integrates sub-network 1, sub-network 2, and the fusion Network, enabling the acquisition of fused HSIs in practical testing under different input conditions through a single training session. The model parameters of the different sub-networks are optimized through the supervision of ground truths (GTs) and the mutual learning strategy. Furthermore, within the network, the MGSAM is designed to extract the spatial edge details at multiple scales sufficiently. The algorithm and network structure are discussed in detail in the following.

### B. Framework

The workflow of the UniNet is shown in Fig. 2, which consists of three parts: sub-network 1, sub-network 2, and the fusion network. $H_{\text{lr}}$, $M_{\text{mr}}$, and $P_{\text{hr}}$ are denoted as the input LR HSI, MR MSI, and HR PAN, respectively. $H_{\text{fuse1}}$, $H_{\text{fuse2}}$, and $H_{\text{fuse3}}$ are denoted as the output HR HSI corresponding to three input conditions. Our goal is to obtain HR HSIs, i.e., $H_{\text{fuse1}}$,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

$H_{\text{fuse2}}$, and $H_{\text{fuse3}}$, which are fused from $H_{\text{lr}}$, $M_{\text{mr}}$, and $P_{\text{hr}}$ in different input conditions via the proposed network

$$\begin{cases} H_{\text{fuse1}} = f_{\text{subnet1}}(H_{\text{lr}}, M_{\text{mr}}) \\ H_{\text{fuse2}} = f_{\text{subnet2}}(H_{\text{lr}}, P_{\text{hr}}) \\ H_{\text{fuse3}} = f_{\text{fusenet}}(H_{\text{fuse1}}, H_{\text{fuse1}}) \end{cases} \tag{1}$$

where $f_{\text{subnet1}}(\cdot)$, $f_{\text{subnet2}}(\cdot)$, and $f_{\text{fusenet}}(\cdot)$ denote the sub-network 1, sub-network 2, and the fusion network, respectively, which are part of the proposed UniNet.

In contrast to previous HIF algorithms that only consider one input condition, our fusion network can achieve HIF under different input conditions. In this way, it is possible to achieve HIF in multiple input conditions via a single model. Next, the various sub-networks of UniNet are described in detail.

*1) Sub-Network 1:* First, $H_{\text{lr}}$ is upscaled to attain the same spatial size as $M_{\text{mr}}$. For upscaled $H_{\text{lr}}$ and $M_{\text{mr}}$, one convolutional layer is applied to obtain fused features initially

$$F_0^1 = f_{\text{conv}_1^1}([H_{\text{lr}} \uparrow, M_{\text{mr}}]) \tag{2}$$

where $\uparrow$ denotes the bicubic interpolation and $f_{\text{conv}_1^1}$ denotes the $1 \times 1$ convolution operation. Then, to extract spatial edge details at multiple scales, $F_0^1$ is passed into $K$ MGSAM

$$F_1^1 = f_{\text{MGSAM}}^K(F_0^1) + F_0^1 \tag{3}$$

where $f_{\text{MGSAM}}^K(\cdot)$ denotes $K$ MGSAMs. To obtain HR HSI, in the final fusion stage, the feature map is passed through one convolutional layer and one deconvolutional layer to obtain $H_{\text{fuse1}}$

$$H_{\text{fuse1}} = f_{\text{up}}\left(f_{\text{conv}_2^1}(F_1^1)\right) + H_{\text{lr}} \uparrow \tag{4}$$

where $f_{\text{up}}$ denotes the upsampling via deconvolution operation. If there is only LR HSI and MR MSI, HIF can be implemented via sub-network 1 to obtain HR HSI.

*2) Sub-Network 2:* First, $H_{\text{lr}}$ is upscaled to obtain the same spatial size as $P_{\text{hr}}$. For upscaled $H_{\text{lr}}$ and $P_{\text{hr}}$, one convolutional layer is applied to obtain fused features initially

$$F_0^2 = f_{\text{conv}_1^2}([H_{\text{lr}} \uparrow, P_{\text{hr}}]) \tag{5}$$

where $f_{\text{conv}_1^2}$ denotes the convolution operation. Afterward, similar to the sub-network 1, $F_0^2$ is passed into $K$ MGSAM to obtain edge details of HSIs and PANs

$$F_1^2 = f_{\text{MGSAM}}^K(F_0^2) + F_0^2. \tag{6}$$

The output feature map $F_1^2$ is then passed through one convolutional layer to finally obtain $H_{\text{fuse2}}$

$$H_{\text{fuse2}} = f_{\text{conv}_2^2}(F_1^2) + H_{\text{lr}} \uparrow . \tag{7}$$

If there is only LR HSI and HR PAN, HIF can be implemented via sub-network 2 to obtain HR HSI.

*3) Fusion Network:* After obtaining $H_{\text{fuse1}}$ and $H_{\text{fuse2}}$, we concatenate the two HR HSIs from sub-network 1 and sub-network 2. The concatenated results are then fed into the fusion network, which consists of a convolutional layer, an MGSAM, and a convolutional layer together. The specific process can be expressed as follows:

$$H_{\text{fuse3}} = f_{\text{conv}_2^3}\left(f_{\text{MGSAM}}^1\left(f_{\text{conv}_1^3}([H_{\text{fuse1}}, H_{\text{fuse2}}])\right)\right) + H_{\text{lr}} \uparrow . \tag{8}$$

If there is LR HSI, MR MSI, and HR PAN simultaneously, HIF can be implemented via the whole network to obtain HR HSI.

Thus, by the proposed UniNet, HIF can be achieved separately under different three input conditions. Specifically, the implementation of HIF is possible via sub-network 1 when HSI and MSI are used as inputs. Furthermore, HIF can be implemented through sub-network 2 when HSI and PAN inputs or through the entire network when all three inputs (i.e., HSI, MSI, and PAN) are available. This "all in one" manner supplies a UniNet for multiple HIF input conditions.

### C. MS Gradient Spatial Attention Module

The extraction of MS features is increasingly crucial in computer vision [40], [41]. Enhancing edge details holds particular significance for low-level tasks [42]. Therefore, extracting spatial edge details at multiple scales presents a highly promising approach for improving image quality. However, the correlation between MS features and edge details has not been further exploited. Thus, we designed the MGSAM to combine the above advantages to improve the quality of fused images via extracting the spatial edge features more comprehensively from multiple scales, as illustrated in Fig. 3. The MS feature extraction process enhances the ability of the network to capture both fine- and coarse-grained features. By learning coarse-and-fine features, the network can effectively extract more continuous spatial features, leading to improved fusion performance. Subsequently, gradient spatial attention (GSA) is a fundamental component of MGSAM, which leverages the spatial gradient information within the feature maps to selectively emphasize regions that contribute more to the quality of fused images. The MGSAM facilitates the network to extract spatial edge details at multiple scales, resulting in more accurate and visually pleasing fused images. The specific details of MGSAM are given in the following.

First, $3 \times 3$, $5 \times 5$, and $7 \times 7$ convolutional layers are used in parallel to obtain MS features from $F_{\text{IN}}$. Afterward, GSA is introduced separately to extract spatial edge details at multiple scales. The specific process of GSA is described as follows:

$$F_{\text{out}} = f_{\text{sigmoid}}(f_{\text{OGM}}(f_{3\times3}(F_{\text{in}}))) \cdot F_{\text{in}} \tag{9}$$

where $f_{3\times3}$ represents a $3 \times 3$ convolution. $f_{\text{OGM}}$ indicates that a pixel-by-pixel subtraction method is used to obtain spatial edge details. $f_{\text{sigmoid}}$ represents the sigmoid function to obtain the spatial edge attention map. $F_{\text{in}}$ and $F_{\text{out}}$ represent the input and output of the GSA. Specifically, in the $3 \times 3$ convolutional layers, the operation of the obtained gradient map (OGM) and the sigmoid function are sequentially introduced to extract spatial edge features by multiplying them with the acquired features. Then, the obtained features from the three branches are concatenated. Two convolutional layers are introduced finally to obtain the output feature maps $F_{\text{OUT}}$. The specific details of MGSAM can be formulated as

$$F_{\text{OUT}} = f_{3\times3}\big(f_{3\times3}\big(\big[f_{\text{GSA}}(f_{3\times3}(F_{\text{IN}})), f_{\text{GSA}}(f_{5\times5}(F_{\text{IN}})) f_{\text{GSA}}(f_{7\times7}(F_{\text{IN}}))\big]\big)\big) \tag{10}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

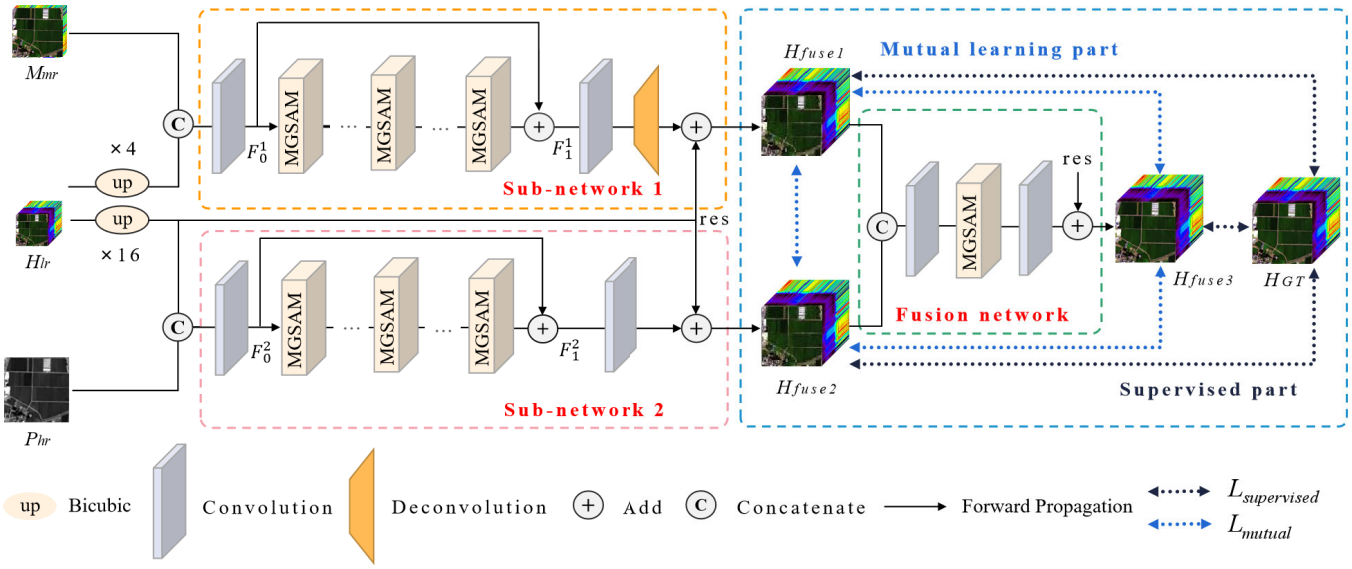



Fig. 2. Architecture of the UniNet. It is possible to fuse HSI with MSI, PAN, or with MSI and PAN via sub-network1, sub-network2, or all sub-networks (i.e., sub-network1, sub-network2, and the fusion network), respectively.

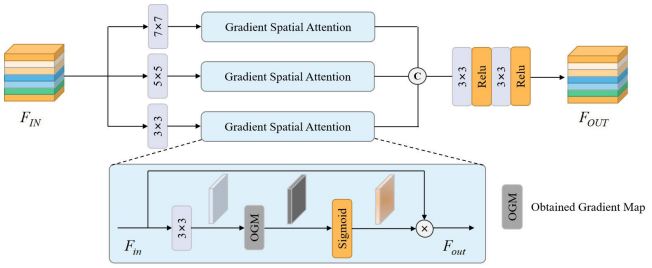



Fig. 3. General flowchart of the MGSAM.

where $f_{5\times5}$ and $f_{7\times7}$ stand for 5 × 5 convolution and 7 × 7 convolution, respectively. $f_{\mathrm{GSA}}$ represents the GSA. The core component of MGSAM, the GSA, allows more attention to be paid to the spatial gradient of the image, thus allowing better reconstruction of the details of HSIs. Meanwhile, the MS network structure combined with GSA can help MGSAM extract richer spatial edge information.

### D. Loss Function With Mutual Learning Strategy

The loss function is essential for optimal network models during the training phase. In the UniNet, a mutual learning strategy is proposed. The mutual learning strategy is effective in the case where the spatial and spectral knowledge experience can be shared between different sub-networks for collaborative learning without direct network connections. The loss function of the proposed UniNet can be divided into a supervised part and a mutual learning part, as shown in Fig. 2. As the proposed network generates three fused images, i.e., $H_{\mathrm{fuse1}}$, $H_{\mathrm{fuse2}}$, and $H_{\mathrm{fuse3}}$ via sub-network 1, sub-network 2, and the fusion network at each training iteration, which is first supervised via GT simultaneously, specifically, we adopt the L1 loss to measure the accuracy of fusion results. This is due to the L1 loss not over-penalizing large errors and exhibiting better convergence performance in contrast to the L2 loss [43]. The L1 loss of $H_{\mathrm{fuse1}}$ is denoted as follows:

$$L_{\mathrm{fuse1}} = \|H_{\mathrm{GT}} - H_{\mathrm{fuse1}}\|_1 \tag{11}$$

where $H_{\mathrm{GT}}$ denotes the GT. Similarly, the supervision of GTs is performed for $H_{\mathrm{fuse2}}$ and $H_{\mathrm{fuse3}}$ simultaneously. Therefore, the loss function for the supervised part is obtained as follows:

$$L_{\mathrm{supervised}} = L_{\mathrm{fuse1}} + L_{\mathrm{fuse2}} + L_{\mathrm{fuse3}}. \tag{12}$$

Because knowledge transfer through direct network connections can be inconvenient for different sub-networks to test separately during the inference phase, we propose the mutual learning strategy for the proposed UniNet, which can prompt different sub-networks to learn and collaborate and does not require direct connection among the networks. The idea of mutual learning was initially proposed for image classification [44]. Unlike the one-way transfer between teacher and student in traditional knowledge distillation, mutual learning is an ensemble manner of students learning together and teaching each other during the training phase. Specifically, for the mutual learning part, we construct the mutual supervision relationship among $H_{\mathrm{fuse1}}$, $H_{\mathrm{fuse2}}$, and $H_{\mathrm{fuse3}}$ by loss function constraints. The HR HSI obtained due to the fusion of different sub-networks possess different characteristics. For example, $H_{\mathrm{fuse1}}$ preserves the spectral correlation better and $H_{\mathrm{fuse2}}$ improves higher spatial resolution. Hence, a mutual learning strategy is proposed, where the spatial and spectral knowledge experience can be shared between different sub-networks for collaborative learning without direct network connections. The loss function for the mutual learning part can be expressed as follows:

$$\begin{aligned} L_{\mathrm{mutual}} = {} & L_1(H_{\mathrm{fuse1}}, H_{\mathrm{fuse2}}) + L_1(H_{\mathrm{fuse2}}, H_{\mathrm{fuse3}}) \\ & + L_1(H_{\mathrm{fuse1}}, H_{\mathrm{fuse3}}) \end{aligned} \tag{13}$$

where $L1(\cdot)$ represents the L1 loss function. The total loss function of the whole network can be expressed as

$$L_{\mathrm{total}} = L_{\mathrm{supervised}} + \lambda * L_{\mathrm{mutual}} \tag{14}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

where $\lambda$ is the balance coefficient. In the practical training phase, we divided into two stages. The first stage utilizes $L_{\text{supervised}}$ and the second stage utilizes $L_{\text{total}}$ as the loss function. This is because adopting a mutual learning strategy at the beginning will lead to more misguided knowledge transfer between networks. This can lead to a decrease in fusion performance. The reason for using the mutual learning strategy in the second stage is that the quality of the fused images is relatively good at this point, and it is easier to learn the best knowledge experiences from each other.

## IV. Experiments

In this section, we perform comprehensive experiments to evaluate the performance of the proposed UniNet. First, four simulated HSI datasets and a real HSI dataset are detailed. Second, the comparison algorithms for HMF, HPF, and HMPF are introduced, respectively. The implementation details and evaluation metrics are then described. Then, the advantages of UniNet are verified by comparing sub-network 1, sub-network 2, and the whole network with state-of-the-art HMF, HPF, and HMPF algorithms, respectively, on each of the four datasets. Next, the spectral preservation of the fused HSI is compared. Finally, the ablation experiments are provided to fully demonstrate the effectiveness of the UniNet.

### A. Datasets

*1) Simulated Datasets:* To verify the advantages of the UniNet, we conducted simulated experiments on Chikusei [45] and Pavia Center [46] datasets. The Chikusei dataset is widely used in the field of hyperspectral remote sensing. This dataset was collected over the Chikusei city in Japan, which consists of 128 spectral bands. In the Chikusei dataset, an intermediate region with a spatial size of 2318 × 2136 is selected, where 80% is divided into the training region and the remaining 20% is divided into the testing region. For the training phase, 96 × 96 overlapping patches are extracted from the training region as GTs. The sizes of HR PAN, MR MSI, and LR HSI patches are 96 × 96, 24 × 24 × 3, and 6 × 6 × 128, respectively. Later, in the inference phase, eight nonoverlapping images are extracted with the size of 256 × 256 × 128 from the testing region. The second selected dataset is the Pavia Center dataset, acquired by the Reflectance Optical System Imaging Spectrometer sensor in an agricultural area of Pavia, Italy. We removed some noisy bands from the Pavia Center dataset and selected 93 bands for the experiment. The top region of the image is selected for training, and the remaining region is selected for testing. To further verify the effectiveness of the UniNet, we conducted experiments on two real-world HSI datasets, i.e., CAVE [47] and Harvard [48]. The CAVE dataset consists of 32 HSIs taken under different lighting conditions. Each HSI in the CAVE dataset contains 31 spectral bands with a spatial size of 512 × 512 pixels. We selected 22 of the CAVE datasets for training and the remaining ten for testing. For the training phase, 128 × 128 overlapping patches are extracted from the training region as GTs. In addition, the sizes of HR PAN, MR MSI, and LR HSI patches are 128 × 128, 36 × 36 × 3, and 9 × 9 × 31, respectively. For the testing phase, the sizes of HR PAN, MR MSI, and LR HSI patches are 512 × 512, 128 × 128 × 3, and 32 × 32 × 31, respectively. In addition, the Harvard dataset consisted of 77 HSIs including outdoor and indoor scenes, with each HSI having a size of 1392 × 1040 × 31. From these, 18 images are chosen, including both outdoor and indoor scenes, and the top-left region (1024 × 1024) is uniformly selected for experiments, using 14 for training and four for testing. The patch size is consistent with the CAVE dataset used for training and testing.

The proposed network requires LR HSIs, MR MSIs, and HR PANs as inputs and HR HSI as GTs for training. Following Wald's protocol [49], the original HSIs of the four datasets are served as GTs. Specifically, the LR HSI is obtained by first applying a Gaussian blurring kernel to blur the GTs and then downsampling the blurred images with a scale factor of 16. The simulated MR MSI is obtained first via Gaussian blurring and downsampling with a scale factor of 4 for GTs and subsequently via integrating the spectral bands using a spectral response function (SRF). Following the settings of [50], SRF of Landset8 is applied to generate MR MSI with three bands from the Chikusei dataset. The IKONOS-like spectral reflectance response filter is applied to generate the MR MSI with four bands from the Pavia Center dataset [51], and the SRF of the Nikon D700 camera is used for CAVE and Harvard datasets to generate the MR MSI with three bands [25]. The simulated PAN with the same spatial size as the GT is obtained by averaging the spectral bands in the visible range of the original HSIs [32]. In this way, GTs, LR HSIs, MR MSIs, and HR PANs are obtained for sufficient HIF experiments to demonstrate the model performance.

*2) Real Datasets:* To further validate the transferability of the algorithm, data from the Hangzhou Bay (HZB) region are collected using the Zi Yuan (ZY)-1 02D satellite, which provided a comprehensive dataset including LR HSI, MR MSI, and HR PAN [52]. The original HSI dataset comprised 166 spectral bands, but, to address low signal-to-noise ratios, only the first 76 bands of the LR HSI are selected for the fusion experiment [52]. As there is no reference image available, the original LR HSI, MR MSI, and HR PAN are spatially downsampled using a Gaussian kernel following Wald's protocol [49] to create the reduced-resolution HZB datasets. Subsequently, the original LR HSI is considered as the reference image. The training subsets are 8 × 8 × 76, 24 × 24 × 8, and 96 × 96 for HSI, MSI, and PAN, while the testing subsets are 32 × 32 × 76, 96 × 96 × 8, and 384 × 384 for HSI, MSI, and PAN, respectively.

### B. Comparison Algorithms

The greatest advantage of UniNet is that it can simultaneously achieve HIF under three input conditions, and all of them can obtain outstanding fusion results. To assess the superiority of the proposed network, we compared the fusion results, i.e., Ours_SR1 ($H_{\text{fuse1}}$), Ours_SR2 ($H_{\text{fuse2}}$), and Ours_SR3 ($H_{\text{fuse3}}$) with results of various state-of-the-art HMF, HPF, and HMPF algorithms, respectively, totaling 18 HIF comparison algorithms.

*1) HMF:* Six HIF algorithms are selected as the baseline for comparison when there is HSI and MSI as input condition:

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

TABLE I
QUANTITATIVE COMPARISON OF THE CHIKUSEI DATASET IN HMF, HPF, AND HMPF TASKS. RED AND GREEN INDICATE THE BEST AND SECOND-BEST VALUES

| Tasks | Metrics | C_HM | LTTR | MSDCNN | SSRNet | Fusformer | PSRT | Ours_SR1 |
|---|---|---|---|---|---|---|---|---|
| HMF | PSNR ↑ | 23.6663 | 24.0704 | 26.4164 | 25.2896 | 26.9430 | 26.6697 | 27.1042 |
| | SAM ↓ | 8.5488 | 8.5245 | 5.7077 | 5.6783 | 5.3848 | 5.2588 | 4.7032 |
| | ERGAS ↓ | 12.8707 | 12.1018 | 9.8813 | 10.1773 | 9.4446 | 9.8232 | 8.5876 |
| | RMSE↓ | 0.0449 | 0.0483 | 0.0325 | 0.0369 | 0.0306 | 0.0317 | 0.0300 |
| | SSIM↑ | 0.8708 | 0.8713 | 0.9338 | 0.9148 | 0.9414 | 0.9377 | 0.9444 |
| | | C_HP | HyperPNN | DRCNN | KitNet | DARN | MDANet | Ours_SR2 |
| HPF | PSNR ↑ | 24.6997 | 23.9310 | 27.0914 | 27.2033 | 28.6206 | 28.8862 | 29.0357 |
| | SAM ↓ | 9.7673 | 9.4962 | 7.9744 | 7.8912 | 7.2038 | 6.5753 | 7.3014 |
| | ERGAS ↓ | 14.8161 | 14.3694 | 13.6569 | 11.7780 | 10.5343 | 10.1589 | 10.0619 |
| | RMSE↓ | 0.0401 | 0.0432 | 0.0301 | 0.0300 | 0.0255 | 0.0251 | 0.0244 |
| | SSIM↑ | 0.8991 | 0.8864 | 0.9450 | 0.9420 | 0.9573 | 0.9572 | 0.9598 |
| | | EXP | C_HMP | CNTR | LTCN | HyperNet | HMPNet | Ours_SR3 |
| HMPF | PSNR ↑ | 22.6273 | 24.4656 | 25.5250 | 23.6960 | 31.3177 | 32.7907 | 33.6565 |
| | SAM ↓ | 9.3897 | 8.7850 | 9.7575 | 8.5815 | 6.2747 | 6.0096 | 4.4538 |
| | ERGAS ↓ | 15.4220 | 12.4272 | 12.3326 | 11.5080 | 7.8136 | 8.0176 | 6.2031 |
| | RMSE↓ | 0.0505 | 0.0410 | 0.0366 | 0.0449 | 0.0186 | 0.0179 | 0.0143 |
| | SSIM↑ | 0.8302 | 0.8951 | 0.9188 | 0.8699 | 0.9787 | 0.9814 | 0.9869 |

CNMF with HSI and MSI as inputs (C_HM) [34], LTTR [19], MS and multidepth CNN (MSDCNN) [53], SSRNet [24], Fusformer [12], and PRST [25]. These include traditional algorithms and deep learning-based algorithms. As the fusion of LR HSI with MR MSI only generates MR HSI, the fusion results obtained from all comparison algorithms are upsampled with a scale factor of 4 through bicubic interpolation to compare with the Ours_SR1 from the sub-network 1 of the UniNet.

*2) HPF:* Six HIF algorithms are selected as the baseline for comparison when there is HSI and PAN as input condition: CNMF with HSI and PAN as inputs (C_HP) [34], HyperPNN [13], deep residual CNN (DRCNN) [54], KitNet [55], dual attention residual network (DARN) [56], and MDANet [14]. These include traditional algorithms and deep learning-based algorithms, and the fused images are compared with the Ours_SR2 from the sub-network 2 of the UniNet.

*3) HMPF:* Six HIF algorithms are selected as the baseline for comparison when there is HSI, MSI, and PAN as input condition: HSI interpolation with a polynomial kernel with 23 coefficients (EXP) [57], CNMF with HSI, MSI and PAN as inputs (C_HMP) [34], CNMF + LTTR (CNTR), LTTR + CNMF (LTCN), HyperNet [37], and HMPNet [16]. These include interpolation algorithms, traditional algorithms, two-step algorithms (HMF + HPF), and deep learning based-algorithms, while the fused images are compared with the Ours_SR3 from the fusion network of the UniNet.

Overall, the relevant hyperparameters of all comparison algorithms are chosen by the original papers and are implemented to achieve the best possible results.

### C. Implementation Details and Evaluation Metrics

In the UniNet, the number of channels for each MGSAM is 96. The numbers of MGSAM are set to 7. In the MGSAM, the number of channels of the feature map is decreased to 16 by $3 \times 3$, $5 \times 5$, and $7 \times 7$ convolutions. Then, the kernel size of the convolutional layer is set to $3 \times 3$ in the GSA. $\lambda$ is set to 0.1. The weights are initialized by the Kaiming initialization [58]. The network is optimized using ADAM [59] with the default setting. The initial learning rate is set at 0.0001 and halved every 50 epochs. The batch size is set at 32. The proposed network is performed on NVIDIA GTX 4090 GPU using the PyTorch framework.

To qualitatively measure the proposed UniNet, five evaluation metrics are used to verify the effectiveness of the proposed network in experiments. Peak signal-to-noise ratio (PSNR) [60] can measure the similarity between the fused image and the GT by calculating the ratio of the maximum possible power of the GT to the mean square error between the fused and GTs. Structural similarity (SSIM) [61] is a metric that measures the similarity between two images by considering the luminance, contrast, and structural information. Higher PSNR and SSIM values indicate a better fusion quality. Spectral angle mapper (SAM) [62] can calculate the angle between the spectral vectors of each pixel in the fused and GTs. Relative dimensionless global error in synthesis (ERGAS) [63] takes into account the spatial and spectral distortions introduced by the fusion process. Root mean square error (RMSE) can measure the average difference between the pixel values of the fused and GTs. A lower SAM, ERGAS, and RMSE value indicates a better fusion quality.

### D. Experimental Results on Remote Sensing HSI Datasets

In this study, we utilized two representative remote sensing datasets to explore the effectiveness of the UniNet, i.e., the Chikusei dataset and the Pavia Center dataset. The quantitative results for the Chikusei dataset are presented in Table I. The best and second-best values are highlighted in red and green, respectively. From the results, Transformer-based algorithms (i.e., Fusformer and PSRT) outperformed CNN-based algorithms (i.e., MCDCNN and SSRNet) and traditional algorithms (i.e., C_HM and LTTR) in terms of HMF. This can be attributed to the Transformer's ability to capture long-range dependencies of pixels. Ours_SR1

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

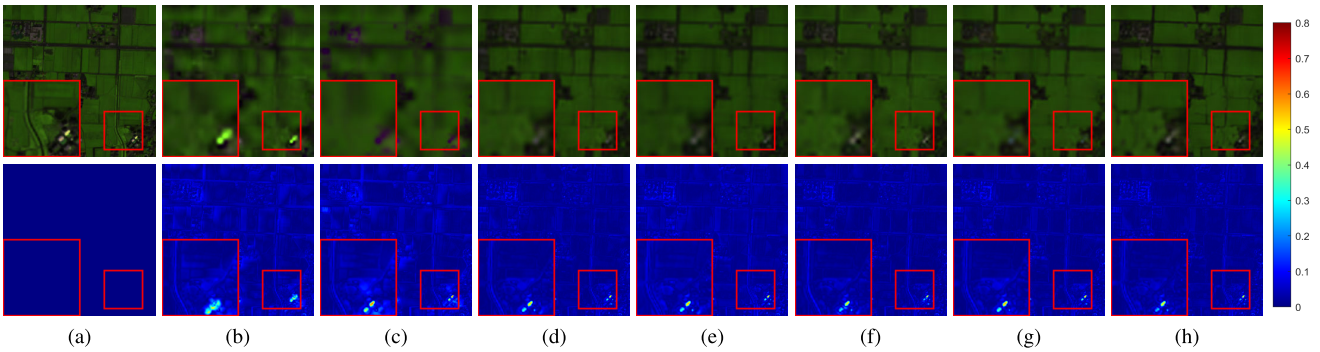



Fig. 4. First row shows the fusion results of HMF using a false color image for clear visualization in the Chikusei dataset. The second row shows the error images between the fused image and GT. (a) GT. (b) C_HM. (c) LTTR. (d) MSDCNN. (e) SSRNet. (f) Fusformer. (g) PSRT. (h) Ours_SR1. The false color image is used combining the 70th, 100th, and 36th bands. The large red box area is obtained by enlarging the small red box area.

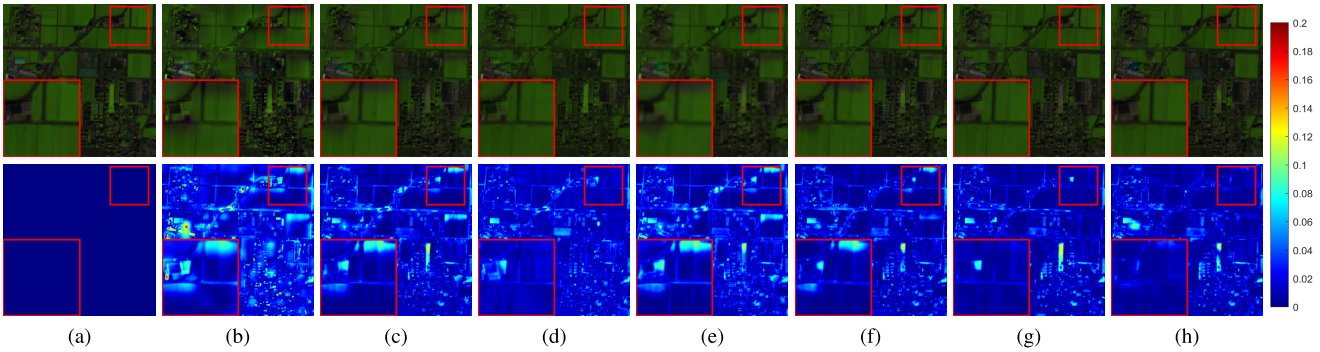



Fig. 5. First row shows the fusion results of HPF using a false color image for clear visualization in the Chikusei dataset. The second row shows the error images between the fused image and GT. (a) GT. (b) C_HP. (c) HyperPNN. (d) DRCNN. (e) KitNet. (f) DARN. (g) MDANet. (h) Ours_SR2. The false color image is used combining the 70th, 100th, and 36th bands. The large red box area is obtained by enlarging the small red box area.

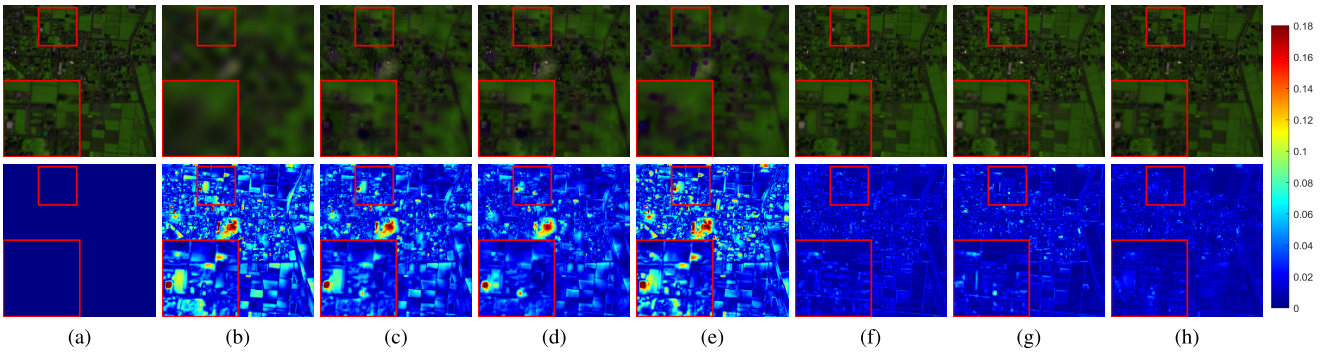



Fig. 6. First row shows the fusion results of HMPF using a false color image for clear visualization in the Chikusei dataset. The second row shows the error images between the fused image and GT. (a) GT. (b) EXP. (c) C_HMP. (d) CNTR. (e) LTCN. (f) HyperNet. (g) HMPNet. (h) Ours_SR3. The false color image is used combining the 70th, 100th, and 36th bands. The large red box area is obtained by enlarging the small red box area.

offers notable benefits compared to the current HMF algorithms. Compared with traditional algorithms i.e., C_HP, deep learning-based algorithms (i.e., HyperPNN, DRCNN, KitNet, DARN, MDANet, and Ours_SR2) show better results in HPF. The UniNet outperforms the other HPF algorithms in all three metrics, i.e., PSNR, ERGAS, and SSIM. Specifically, PSNR/ERGAS/SSIM exhibits an enhancement of 0.1495/0.0970/0.0017 in comparison to the second-best outcomes (i.e., MDANet). Ours_SR3 also shows obvious advantages compared to recent deep learning-based algorithms (i.e., HyperNet and HMPNet). In addition, qualitative assessments on the Chikusei dataset are shown in Figs. 4–6, which include fused images and absolute error images. For better representation, the fused images are illustrated using false color images by combining the 70th, 100th, and 36th bands. The error image becomes more bluish as the fused image improves. Fig. 4 illustrates that Ours_SR1 exhibits sharper edge details compared to other HMF algorithms. The error images of Ours_SR2 and Ours_SR3 in Figs. 5 and 6 display a smaller error compared to HPF and HMPF algorithms, indicating that our fused image is closer to GT.

The quantitative and qualitative results of the Pavia Center dataset are shown in Table II and Figs. 7–9. Almost all metrics of Ours_SR1, Ours_SR2, and Ours_SR3 perform the best compared to the other algorithms. In HMF, an improvement of 1.0717/0.8614/0.0335 in PSNR/ERGAS/SSIM of Ours_SR1

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

TABLE II

QUANTITATIVE COMPARISON OF THE PAVIA CENTER DATASET IN HMF, HPF, AND HMPF TASKS. RED AND GREEN INDICATE THE BEST AND SECOND-BEST VALUES

| Tasks | Metrics | C_HM | LTTR | MSDCNN | SSRNet | Fusformer | PSRT | Ours_SR1 |
|---|---|---|---|---|---|---|---|---|
| HMF | PSNR ↑ | 22.9303 | 23.2816 | 26.0053 | 25.2676 | 26.5388 | 27.1611 | 28.2328 |
| | SAM ↓ | 9.2773 | 8.8218 | 6.8809 | 8.2526 | 5.8929 | 5.7199 | 5.2275 |
| | ERGAS ↓ | 11.7447 | 11.4944 | 8.2410 | 8.5772 | 7.7739 | 7.2661 | 6.4047 |
| | RMSE↓ | 0.1112 | 0.1068 | 0.0780 | 0.0849 | 0.0733 | 0.0683 | 0.0603 |
| | SSIM↑ | 0.6614 | 0.7603 | 0.8213 | 0.7918 | 0.8500 | 0.8685 | 0.9020 |
| | | C_HP | HyperPNN | DRCNN | KitNet | DARN | MDANet | Ours_SR2 |
| HPF | PSNR ↑ | 24.8536 | 27.7389 | 28.1703 | 29.0701 | 29.8345 | 30.100 | 29.9870 |
| | SAM ↓ | 11.0738 | 9.8225 | 10.6649 | 8.5165 | 7.7236 | 7.3396 | 8.0309 |
| | ERGAS ↓ | 9.2140 | 6.7075 | 6.3978 | 5.6785 | 5.2735 | 5.2700 | 5.2293 |
| | RMSE↓ | 0.0895 | 0.0645 | 0.0607 | 0.0548 | 0.0501 | 0.0486 | 0.0495 |
| | SSIM↑ | 0.7938 | 0.9067 | 0.9072 | 0.9203 | 0.9370 | 0.9379 | 0.9380 |
| | | EXP | C_HMP | CNTR | LTCN | HyperNet | HMPNet | Ours_SR3 |
| HMPF | PSNR ↑ | 20.0937 | 24.6193 | 25.3779 | 25.6585 | 34.7711 | 36.7769 | 36.8992 |
| | SAM ↓ | 10.2593 | 9.9867 | 11.3934 | 11.2003 | 4.7503 | 4.0238 | 3.8027 |
| | ERGAS ↓ | 14.5447 | 9.5959 | 8.7884 | 9.8876 | 3.0512 | 2.4443 | 2.4191 |
| | RMSE↓ | 0.1374 | 0.0916 | 0.0839 | 0.0912 | 0.0284 | 0.0226 | 0.0223 |
| | SSIM↑ | 0.4113 | 0.7935 | 0.8340 | 0.8546 | 0.9800 | 0.9874 | 0.9878 |

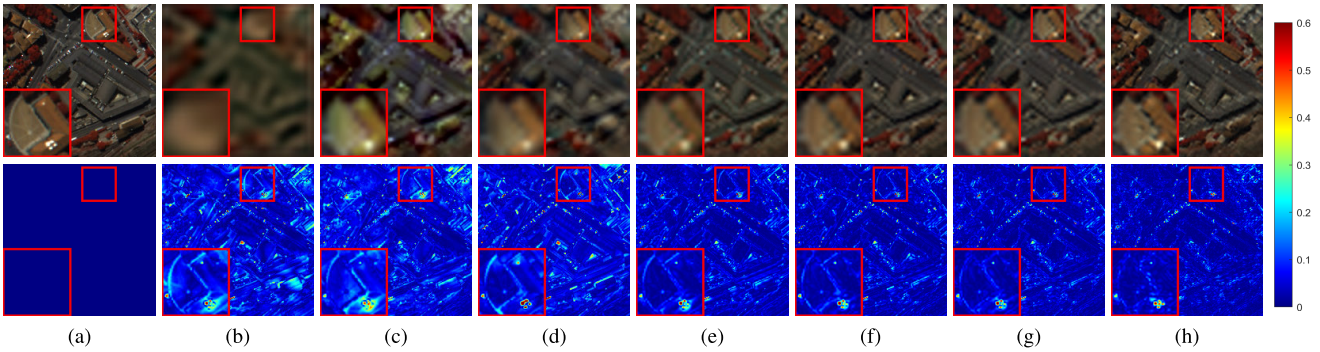



Fig. 7. First row shows the fusion results of HMF using a false color image for clear visualization in the Pavia Center dataset. The second row shows the error images between the fused image and GT. (a) GT. (b) C_HM. (c) LTTR. (d) MSDCNN. (e) SSRNet. (f) Fusformer. (g) PSRT. (h) Ours_SR1. The false color image is used combining the 64th, 32nd, and 11th bands. The large red box area is obtained by enlarging the small red box area.

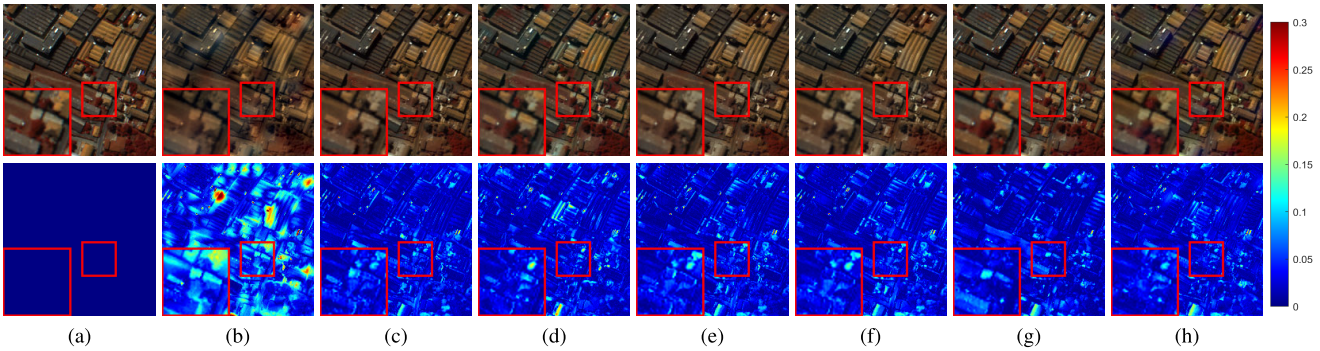



Fig. 8. First row shows the fusion results of HPF using a false color image for clear visualization in the Pavia Center dataset. The second row shows the error images between the fused image and GT. (a) GT. (b) C_HP. (c) HyperPNN. (d) DRCNN. (e) KitNet. (f) DARN. (g) MDANet. (h) Ours_SR2. The false color image is used combining the 64th, 32nd, and 11th bands. The large red box area is obtained by enlarging the small red box area.

compared with the second-best algorithm (i.e., PSRT). In HPF, Ours_SR2 is the best on two metrics (i.e., ERGAS and SSIM) and the second-best on PSNR, which proves the superiority of UniNet. In HMPF, an improvement of 0.1223/0.0252/0.0004 in PSNR/ERGAS/SSIM of Ours_SR3 is obtained compared with the second-best algorithm (i.e., HMPNet). For better representation, the fused images are illustrated using false color images by combining the 64th, 32nd, and 11th bands. In Fig. 7, it can be seen that Ours_SR1 shows a sharper edge contour compared to the other fused images. In Figs. 8 and 9, Ours_SR2 is closest to GT in terms of roof color, and Ours_SR3 presents smaller absolute error compared to the other HMPF images.

### *E. Experimental Results on Real-world HSI Datasets*

In the pursuit of real-world scenes, we carefully curated two datasets (i.e., CAVE and Harvard datasets) for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

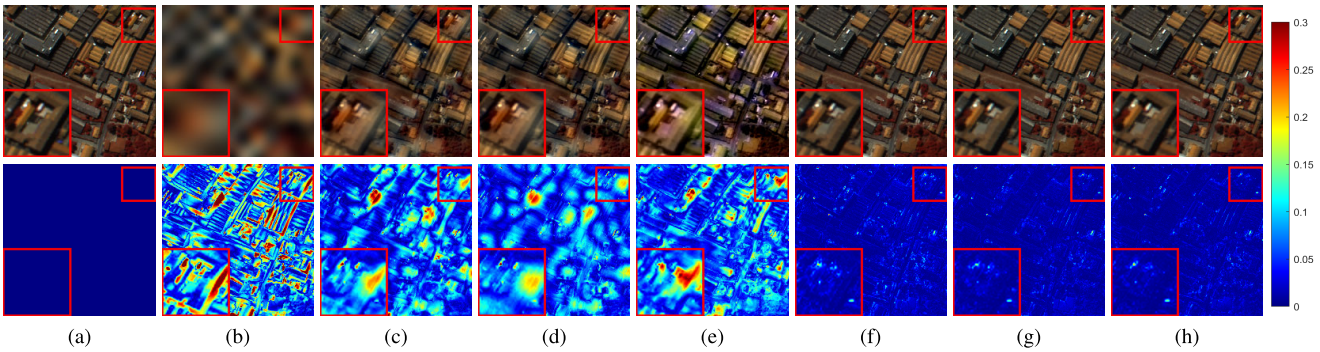



Fig. 9. First row shows the fusion results of HMPF using a false color image for clear visualization in the Pavia Center dataset. The second row shows the error images between the fused image and GT. (a) GT. (b) EXP. (c) C_HMP. (d) CNTR. (e) LTCN. (f) HyperNet. (g) HMPNet. (h) Ours_SR3. The false color image is used combining the 64th, 32nd, and 11th bands. The large red box area is obtained by enlarging the small red box area.

TABLE III

QUANTITATIVE COMPARISON OF THE CAVE DATASET IN HMF, HPF, AND HMPF TASKS. RED AND GREEN INDICATE THE BEST AND SECOND-BEST VALUES

| Tasks | Metrics | C_HM | LTTR | MSDCNN | SSRNet | Fusformer | PSRT | Ours_SR1 |
|---|---|---|---|---|---|---|---|---|
| | PSNR ↑ | 26.2034 | 28.2462 | 30.5627 | 31.4720 | 32.0016 | 32.4276 | 34.1483 |
| | SAM ↓ | 9.2731 | 10.5013 | 8.1032 | 7.3026 | 6.8731 | 6.5026 | 6.0473 |
| HMF | ERGAS ↓ | 11.3850 | 8.4680 | 6.6919 | 6.0178 | 5.8758 | 5.6680 | 4.6737 |
| | RMSE↓ | 0.0505 | 0.0401 | 0.0308 | 0.0275 | 0.0260 | 0.0254 | 0.0206 |
| | SSIM↑ | 0.9164 | 0.9486 | 0.9699 | 0.9734 | 0.9752 | 0.9739 | 0.9832 |
| | | C_HP | HyperPNN | DRCNN | KitNet | DARN | MDANet | Ours_SR2 |
| | PSNR ↑ | 25.5325 | 29.2136 | 30.9997 | 32.8393 | 33.8607 | 34.0122 | 34.0438 |
| | SAM ↓ | 9.7633 | 12.3304 | 8.3260 | 7.0803 | 6.5557 | 6.8615 | 6.6361 |
| HPF | ERGAS ↓ | 12.9013 | 8.8821 | 6.6131 | 4.8018 | 4.4478 | 4.4644 | 4.4089 |
| | RMSE↓ | 0.0541 | 0.0358 | 0.0286 | 0.0248 | 0.0215 | 0.0213 | 0.0211 |
| | SSIM↑ | 0.9238 | 0.9475 | 0.9745 | 0.9791 | 0.9842 | 0.9837 | 0.9842 |
| | | EXP | C_HMP | CNTR | LTCN | HyperNet | HMPNet | Ours_SR3 |
| | PSNR ↑ | 20.7690 | 29.8339 | 30.0024 | 31.9354 | 36.1368 | 36.2406 | 37.2912 |
| | SAM ↓ | 12.6920 | 9.4109 | 11.6809 | 9.3156 | 5.6612 | 5.3756 | 5.2354 |
| HMPF | ERGAS↓ | 20.5548 | 7.3891 | 7.3415 | 5.0893 | 3.1900 | 3.7035 | 2.8972 |
| | RMSE ↓ | 0.0955 | 0.0331 | 0.0355 | 0.0262 | 0.0162 | 0.0142 | 0.0142 |
| | SSIM↑ | 0.7955 | 0.9562 | 0.9627 | 0.9752 | 0.9899 | 0.9899 | 0.9918 |

TABLE IV

QUANTITATIVE COMPARISON OF THE HARVARD DATASET IN HMF, HPF, AND HMPF TASKS. RED AND GREEN INDICATE THE BEST AND SECOND-BEST VALUES

| Tasks | Metrics | C_HM | LTTR | MSDCNN | SSRNet | Fusformer | PSRT | Ours_SR1 |
|---|---|---|---|---|---|---|---|---|
| | PSNR ↑ | 32.4146 | 32.3164 | 34.3339 | 34.7673 | 35.5345 | 35.8684 | 37.7490 |
| | SAM ↓ | 4.5551 | 5.7300 | 4.7588 | 4.6983 | 4.5341 | 4.4569 | 4.1103 |
| HMF | ERGAS ↓ | 6.4673 | 7.0210 | 5.2004 | 6.0268 | 5.0056 | 5.3601 | 6.1435 |
| | RMSE↓ | 0.0330 | 0.0314 | 0.0234 | 0.0243 | 0.0229 | 0.0238 | 0.0183 |
| | SSIM↑ | 0.9473 | 0.9569 | 0.9755 | 0.9731 | 0.9752 | 0.9742 | 0.9835 |
| | | C_HP | HyperPNN | DRCNN | KitNet | DARN | MDANet | Ours_SR2 |
| | PSNR ↑ | 33.9225 | 35.9488 | 39.7738 | 40.1023 | 40.3985 | 41.9041 | 42.3757 |
| | SAM ↓ | 4.6702 | 6.8050 | 4.6091 | 4.5614 | 4.4850 | 4.3074 | 4.2667 |
| HPF | ERGAS ↓ | 5.8734 | 7.7054 | 3.9933 | 3.6060 | 3.9566 | 3.3072 | 3.2969 |
| | RMSE↓ | 0.0271 | 0.0191 | 0.0126 | 0.0124 | 0.0117 | 0.0099 | 0.0095 |
| | SSIM↑ | 0.9650 | 0.9817 | 0.9921 | 0.9928 | 0.9941 | 0.9954 | 0.9957 |
| | | EXP | C_HMP | CNTR | LTCN | HyperNet | HMPNet | Ours_SR3 |
| | PSNR ↑ | 26.3710 | 36.6515 | 37.6613 | 37.6010 | 41.7141 | 42.1113 | 43.3677 |
| | SAM ↓ | 5.1163 | 4.3903 | 5.0767 | 5.7090 | 4.2595 | 4.3793 | 3.9291 |
| HMPF | ERGAS ↓ | 11.7263 | 4.4934 | 4.7288 | 5.1119 | 8.9736 | 3.3527 | 2.8302 |
| | RMSE↓ | 0.0638 | 0.0199 | 0.0127 | 0.0215 | 0.0106 | 0.0102 | 0.0085 |
| | SSIM↑ | 0.8509 | 0.9788 | 0.9907 | 0.9828 | 0.9961 | 0.9961 | 0.9968 |

evaluation. For the CAVE dataset, the best quantitative results are obtained with Ours_SR1, Ours_SR2, and Ours_SR3, as shown in Table III. Specifically, an improvement of 1.7207/0.9943/0.0093 in PSNR/ERGAS/SSIM was achieved for HMF compared to the second-best results (i.e., PSRT). In Fig. 10, the shape of the object details, as well as the edges,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

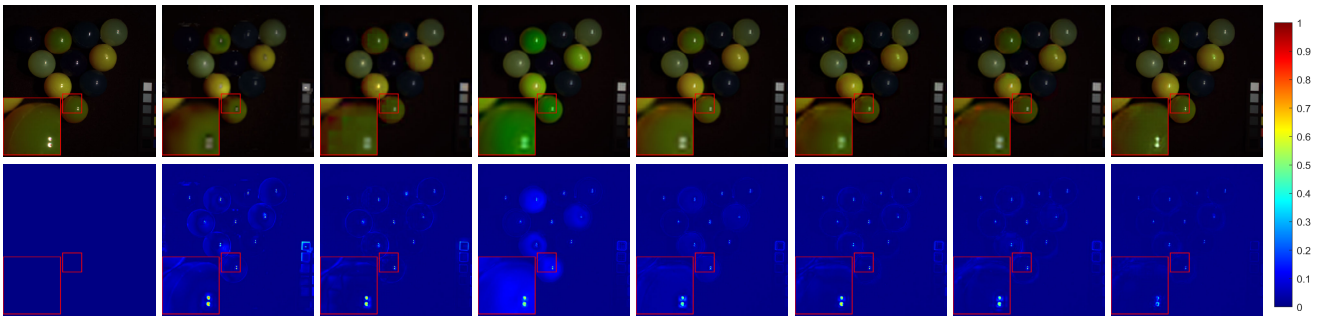



Fig. 10. First row shows the fusion results of HMF using a false color image for clear visualization in the CAVE dataset. The second row shows the error images between the fused image and GT. (a) GT. (b) C_HM. (c) LTTR. (d) MSDCNN. (e) SSRNet. (f) Fusformer. (g) PSRT. (h) Ours_SR1. The false color image is used combining the 30th, 20th, and tenth bands. The large red box area is obtained by enlarging the small red box area.

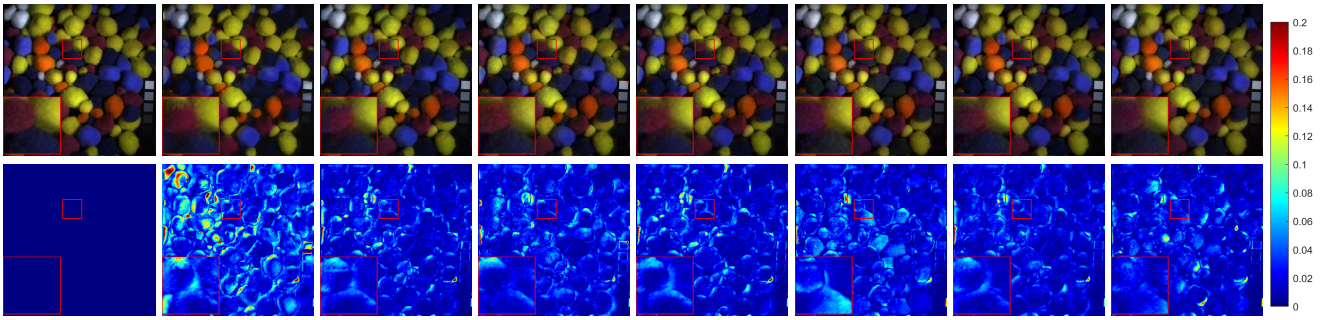



Fig. 11. First row shows the fusion results of HPF using a false color image for clear visualization in the CAVE dataset. The second row shows the error images between the fused image and GT. (a) GT. (b) C_HP. (c) HyperPNN. (d) DRCNN. (e) KitNet. (f) DARN. (g) MDANet. (h) Ours_SR2. The false color image is used combining the 30th, 20th, and tenth bands. The large red box area is obtained by enlarging the small red box area.

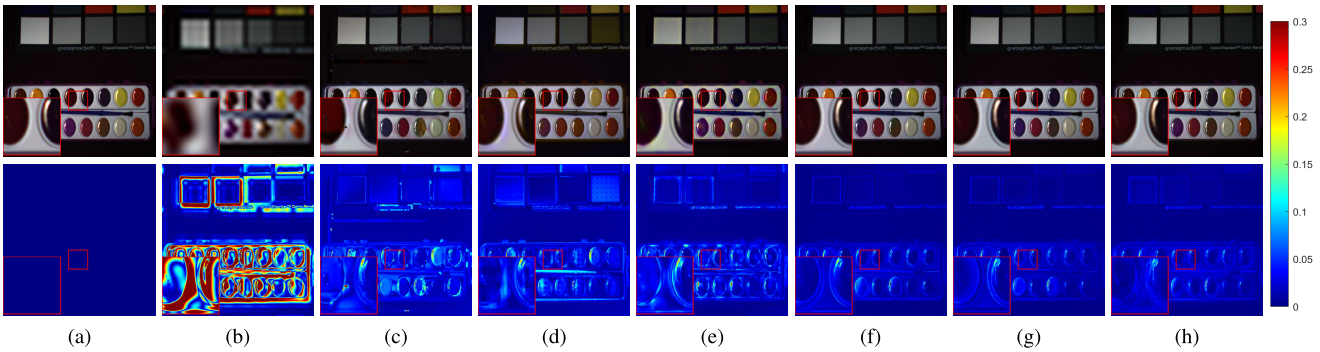



Fig. 12. First row shows the fusion results of HMPF using a false color image for clear visualization in the CAVE dataset. The second row shows the error images between the fused image and GT. (a) GT. (b) EXP. (c) C_HMP. (d) CNTR. (e) LTCN. (f) HyperNet. (g) HMPNet. (h) Ours_SR3. The false color image is used combining the 30th, 20th, and tenth bands. The large red box area is obtained by enlarging the small red box area.

is more similar to GT. This is attributed to the knowledge that Ours_SR1 can receive high spatial resolution information from Ours_SR2 and Ours_SR3 via the mutual learning strategy. It is notable that for better representation, the fused images are illustrated using false color images by combining the 30th, 20th, and tenth bands. From Table III, it can be also observed that Ours_SR2 outperforms the existing HPF algorithms on all three metrics. The error images show that the gap between the Ours_SR2 and GT is minimal in Fig. 11. In addition, Ours_SR3 also outperforms the existing HMPF in all three metrics, indicating that UniNet shows an advantage compared to the state-of-the-art HMPF algorithms. The visual result of Ours_SR3 in Fig. 12 is also optimal. These further demonstrate the advantages of the UniNet in real-world datasets.

With the Harvard dataset, the best performance is obtained for Ours_SR1, Ours_SR2, and Ours_SR3, as shown in Table IV. As for HMF, the PSNR of Ours_SR1 is 1.8806 dB higher than that of PRST. For better representation, the fused images are illustrated using false color images by combining the 30th, 20th, and tenth bands. The absolute error images in Fig. 13 show that Ours_SR1 is the closest to GT. This indicates that our UniNet is superior in spatial reconstruction and spectral preservation. Moreover, the UniNet is also advantageous for HPF and HMPF, achieving the best metrics and the greatest visuals, as shown in Table IV and Figs. 14 and 15.

### *F. Results of Spectral Preservation*

To compare the ability of the UniNet to preserve the spectral correlation of HSIs with state-of-the-art algorithms, as depicted in Fig. 16, possible spectral distortions in the fused images are illustrated by showing the spectral curves. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

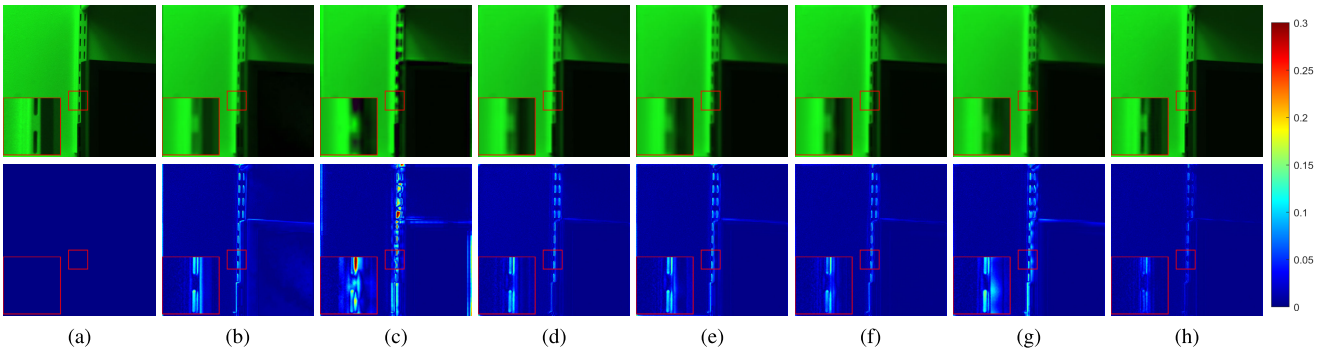



Fig. 13. First row shows the fusion results of HMF using a false color image for clear visualization in the Harvard dataset. The second row shows the error images between the fused image and GT. (a) GT. (b) C_HM. (c) LTTR. (d) MSDCNN. (e) SSRNet. (f) Fusformer. (g) PSRT. (h) Ours_SR1. The false color image is used combining the 30th, 20th, and tenth bands. The large red box area is obtained by enlarging the small red box area.

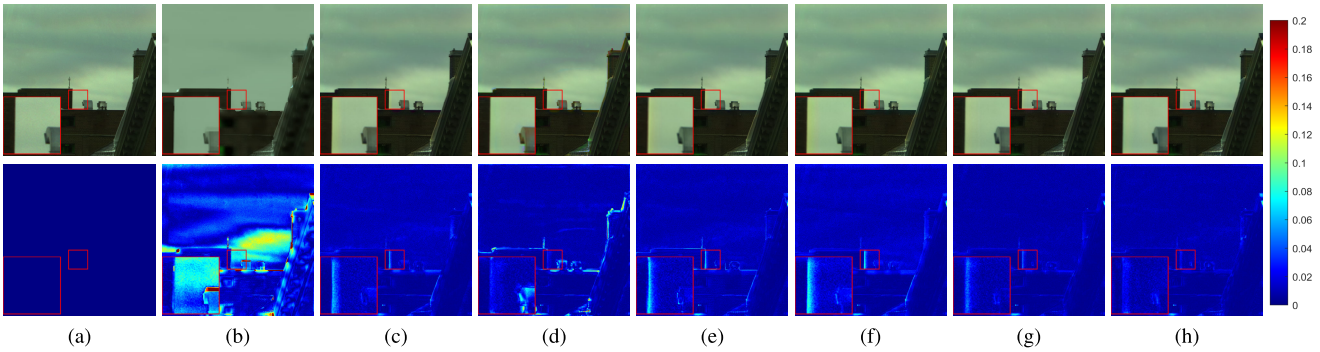



Fig. 14. First row shows the fusion results of HPF using a false color image for clear visualization in the Harvard dataset. The second row shows the error images between the fused image and GT. (a) GT. (b) C_HP. (c) HyperPNN. (d) DRCNN. (e) KitNet. (f) DARN. (g) MDANet. (h) Ours_SR2. The false color image is used combining the 30th, 20th, and tenth bands. The large red box area is obtained by enlarging the small red box area.

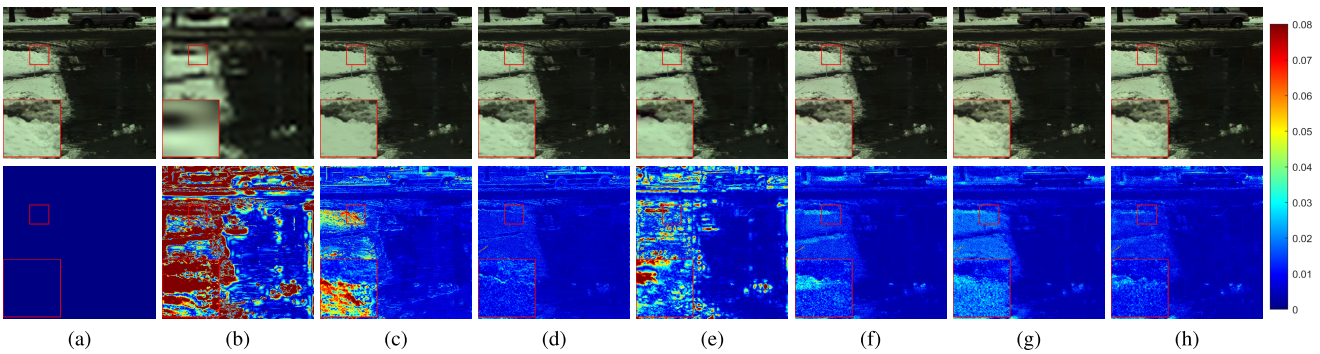



Fig. 15. First row shows the fusion results of HMPF using a false color image for clear visualization in the Harvard dataset. The second row shows the error images between the fused image and GT. (a) GT. (b) EXP. (c) C_HMP. (d) CNTR. (e) LTCN. (f) HyperNet. (g) HMPNet. (h) Ours_SR3. The false color image is used combining the 30th, 20th, and tenth bands. The large red box area is obtained by enlarging the small red box area.

pixel is selected randomly from the test image to compare the spectral similarity between GT and fused image in the Chikusei, Pavia Center, CAVE, and Harvard datasets. Spectral curves can reflect the values of different spectral bands in HSI. It can be seen that the fused images obtained by the UniNet are all closest to GT in spectral vectors under different input conditions. This is a significant indication that UniNet can better conquer spectral distortion, which is especially essential in the HIF task.

### G. Experiments on Real HSI Datasets

To further highlight the effectiveness of the proposed method, experiments on real data are conducted on the HZB dataset, and the fusion results are presented in Fig. 17. The well-trained model, initially trained on the reduced-resolution dataset, is directly used for the original image fusion to obtain results on real HSI data. Among the results in Fig. 17(a)–(g), Ours_SR1 demonstrates the most superior contour detail recovery. Similarly, as observed in (h)–(n), Ours_SR2 excels in clear detail reconstruction. In addition, in (o)–(u), Ours_SR3 shows the best spatial detail restoration. These results reinforce the efficacy of the proposed method on real data, amplifying its practical utility and applicability.

### H. Ablation Experiments

A series of ablation experiments are performed on the Harvard dataset to verify the effectiveness of the proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

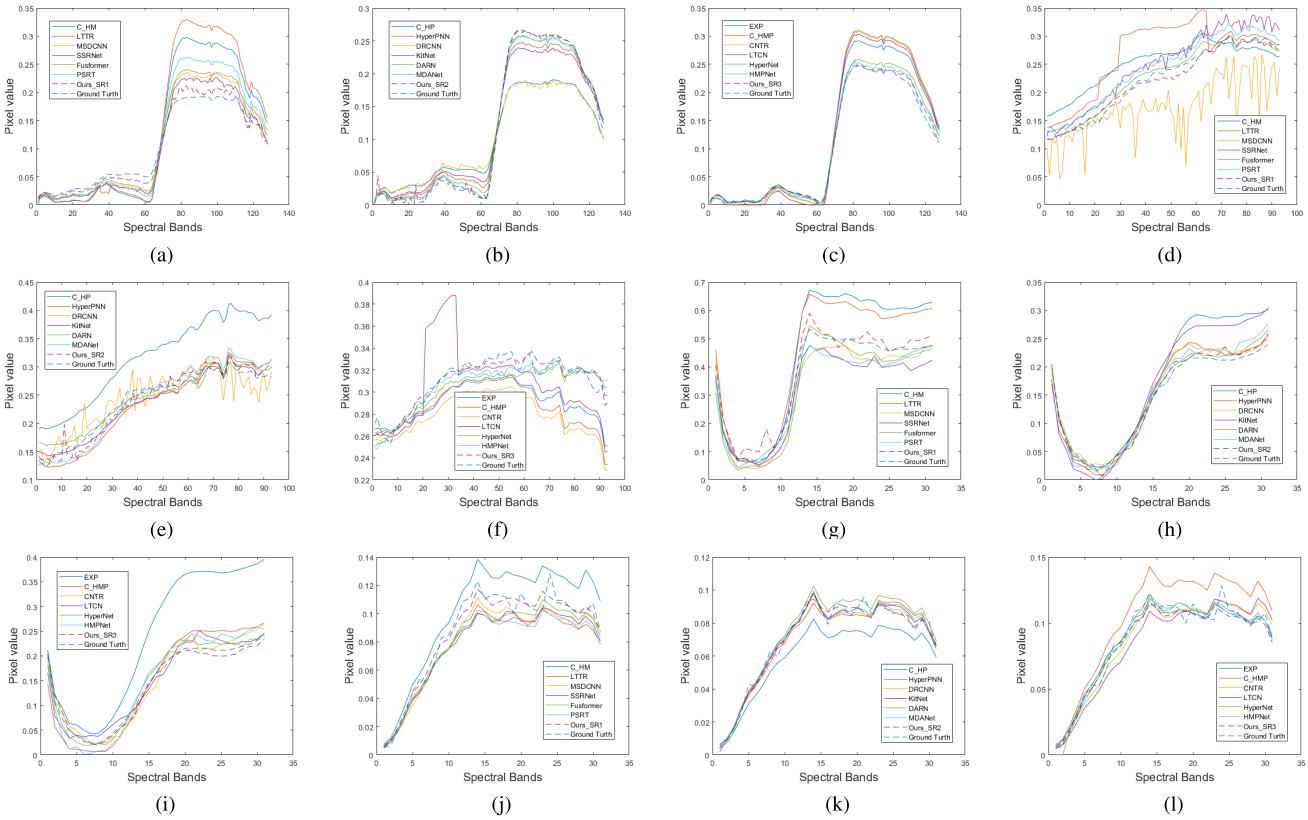



Fig. 16. Spectral curves of the UniNet and the existing HIF algorithms in three input conditions (i.e., HMF, HPF, and HMPF) with four public datasets. (a)–(c) HMF, HPF, and HMPF on the Chikusei dataset. (d)–(f) HMF, HPF, and HMPF on the Pavia Center dataset. (g)–(i) HMF, HPF, and HMPF on the CAVE dataset. (j)–(l) HMF, HPF, and HMPF on the Harvard dataset.

TABLE V

STUDY ON THE NUMBER OF MGSAMS IN UNINET ON THE HARVARD DATASETS. RED INDICATES THE BEST VALUE

| | Metrics | 5 | 6 | 7 (Ours) | 8 |
|---|---|---|---|---|---|
| HMF | PSNR ↑ | 37.4697 | 37.5305 | 37.7490 | 37.5856 |
| | SAM ↓ | 4.2870 | 4.2532 | 4.1103 | 4.2698 |
| | ERGAS ↓ | 12.6364 | 12.5359 | 10.1435 | 11.8721 |
| | RMSE↓ | 0.0187 | 0.0187 | 0.0183 | 0.0186 |
| | SSIM↑ | 0.9830 | 0.9831 | 0.9835 | 0.9831 |
| HPF | PSNR ↑ | 42.0553 | 42.0542 | 42.3757 | 42.2205 |
| | SAM ↓ | 4.4482 | 4.4220 | 4.2667 | 4.4026 |
| | ERGAS ↓ | 3.5148 | 3.4319 | 3.2969 | 3.4056 |
| | RMSE↓ | 0.0097 | 0.0098 | 0.0095 | 0.0096 |
| | SSIM↑ | 0.9955 | 0.9955 | 0.9957 | 0.9956 |
| HMPF | PSNR ↑ | 42.9820 | 42.8831 | 43.3677 | 42.5041 |
| | SAM ↓ | 3.9660 | 4.0549 | 3.9291 | 4.2115 |
| | ERGAS ↓ | 2.9627 | 2.9971 | 2.8302 | 2.9422 |
| | RMSE↓ | 0.0088 | 0.0088 | 0.0085 | 0.0091 |
| | SSIM↑ | 0.9966 | 0.9966 | 0.9968 | 0.9961 |

TABLE VI

ABLATION STUDY WITH MS AND GSA ON THE HARVARD DATASET. RED INDICATES THE BEST VALUE

| MS | GSA | Tasks | PSNR↑ / SAM↓ / ERGAS ↓ / RMSE↓ / SSIM↑ |
|---|---|---|---|
| √ | × | HMF | 37.5176 / 4.3023 / 13.0847 / 0.0186 / 0.9832 |
| | | HPF | 42.1717 / 4.3988 / 3.3981 / 0.0096 / 0.9956 |
| | | HMPF | 42.8008 / 4.0638 / 3.0109 / 0.0091 / 0.9966 |
| × | √ | HMF | 37.4896 / 4.3198 / 12.4454 / 0.0187 / 0.9831 |
| | | HPF | 41.8483 / 4.4877 / 3.4818 / 0.0100 / 0.9953 |
| | | HMPF | 42.7964 / 4.0608 / 2.9317 / 0.0089 / 0.9966 |
| √ | √ | HMF | 37.7490 / 4.1103 / 10.1435 / 0.0183 / 0.9835 |
| | | HPF | 42.3757 / 4.2667 / 3.2969 / 0.0095 / 0.9957 |
| | | HMPF | 43.3677 / 3.9291 / 2.8302 / 0.0085 / 0.9968 |

UniNet. First, the number of MGSAMs is set from 5 to 8 to analyze the parameter $K$. Table V shows the results of the five metrics in three input conditions, which indicates that a larger $K$ shows better performance when $K$ is less than 7. Excellent fusion performance is possible with deeper networks. Nevertheless, when $K$ is 8, there is a decrease in the evaluation metrics. The deeper the network, the more difficult the training is, and this is the main reason for this phenomenon. For this reason, in the proposed network, $K$ is set to 7.

To further validate the advantages of the MGSAM in the UniNet, we constructed ablation experiments on MS and GSA. The experimental results are shown in Table VI. The results show that PSNR significantly decreases after removing GSA because GSA is more effective at the extraction of spatial edge details. When MS (i.e., $3 \times 3$, $5 \times 5$, and $7 \times 7$) is replaced with single-scale (i.e., $3 \times 3$, $3 \times 3$, and $3 \times 3$), all five metrics show a decrease simultaneously. This is possible because convolutions of different sizes can provide multiple scale features, thus contributing to more continuous information

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
ABLATION STUDY WITH MUTUAL LEARNING STRATEGY ON THE HARVARD DATASET. RED INDICATES THE BEST VALUE

| | Metrics | Single stage | $\lambda$=0 | $\lambda$=0.1(Ours) | $\lambda$=0.2 | $\lambda$=0.3 | $\lambda$=0.4 | $\lambda$=0.5 | $\lambda$=0.6 | $\lambda$=0.7 | $\lambda$=0.8 | $\lambda$=0.9 | $\lambda$=1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| HMF | PSNR ↑ | 37.6085 | 37.5908 | 37.7490 | 37.5691 | 37.5712 | 37.3809 | 37.4084 | 37.3773 | 37.3132 | 37.2225 | 37.0886 | 37.0523 |
| | SAM ↓ | 4.3072 | 4.3429 | 4.1103 | 4.2822 | 4.2984 | 4.4129 | 4.4665 | 4.4326 | 4.4998 | 4.5299 | 4.5529 | 4.6098 |
| | ERGAS ↓ | 12.1384 | 12.687 | 10.1435 | 12.7206 | 12.8899 | 13.4718 | 13.3732 | 13.4531 | 13.7637 | 14.0306 | 13.9927 | 14.4890 |
| | RMSE↓ | 0.0185 | 0.0186 | 0.0183 | 0.0186 | 0.0185 | 0.0189 | 0.0188 | 0.0189 | 0.0189 | 0.0191 | 0.0193 | 0.0194 |
| | SSIM↑ | 0.9833 | 0.9831 | 0.9835 | 0.9831 | 0.9832 | 0.9826 | 0.9828 | 0.9827 | 0.9826 | 0.9823 | 0.9819 | 0.9818 |
| HPF | PSNR ↑ | 41.7515 | 42.0629 | 42.3757 | 41.9682 | 41.9562 | 41.7710 | 41.7393 | 41.3883 | 41.1784 | 40.8899 | 40.4096 | 39.8301 |
| | SAM ↓ | 4.5109 | 4.5375 | 4.2667 | 4.4520 | 4.4469 | 4.4009 | 4.4235 | 4.4741 | 4.5416 | 4.4912 | 4.4943 | 4.6418 |
| | ERGAS ↓ | 3.5341 | 3.5448 | 3.2969 | 3.4791 | 3.5282 | 3.4927 | 3.6544 | 3.6903 | 3.9174 | 4.3133 | 4.8911 | 5.6613 |
| | RMSE↓ | 0.0101 | 0.0097 | 0.0095 | 0.0098 | 0.0099 | 0.0100 | 0.0101 | 0.0105 | 0.0108 | 0.0112 | 0.0119 | 0.0129 |
| | SSIM↑ | 0.9955 | 0.9954 | 0.9957 | 0.9955 | 0.9954 | 0.9952 | 0.9951 | 0.9947 | 0.9944 | 0.9939 | 0.9931 | 0.9920 |
| HMPF | PSNR ↑ | 42.0467 | 42.9142 | 43.3677 | 42.6775 | 42.5550 | 42.3205 | 41.9478 | 41.6021 | 41.3373 | 40.8167 | 40.3024 | 39.7799 |
| | SAM ↓ | 4.1106 | 4.0196 | 3.9291 | 4.0214 | 4.0981 | 4.1358 | 4.1502 | 4.1535 | 4.1983 | 4.2793 | 4.3401 | 4.4237 |
| | ERGAS ↓ | 3.1144 | 3.0517 | 2.8302 | 2.9531 | 3.2013 | 3.1052 | 3.1739 | 3.3702 | 3.5006 | 3.9576 | 4.8753 | 5.4764 |
| | RMSE↓ | 0.0102 | 0.0090 | 0.0085 | 0.0092 | 0.0092 | 0.0094 | 0.0099 | 0.0103 | 0.0106 | 0.0113 | 0.0121 | 0.0131 |
| | SSIM↑ | 0.9964 | 0.9967 | 0.9968 | 0.9964 | 0.9962 | 0.9960 | 0.9957 | 0.9953 | 0.9949 | 0.9941 | 0.9931 | 0.9919 |

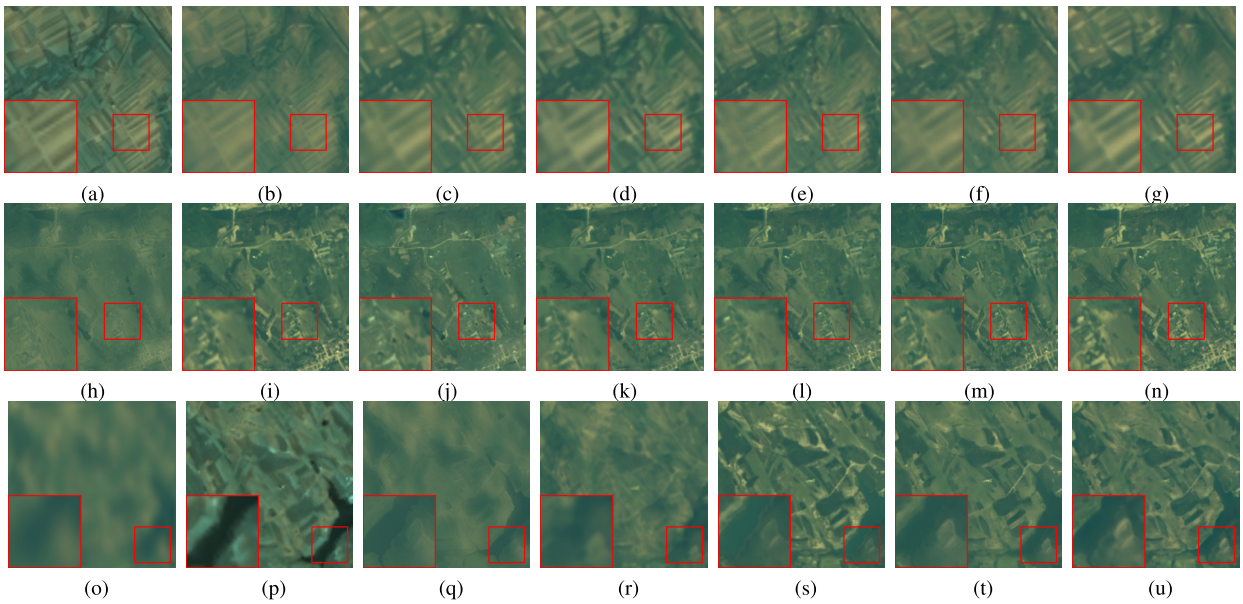



Fig. 17. Real HSI fusion results in the HZB dataset. (a)–(g) Results of HM. (h)–(n) Results of HM. (o)–(u) Results of HM. (a) C_HM. (b) LTTR. (c) MSDCNN. (d) SSRNet. (e) Fusformer. (f) PSRT. (g) Ours_SR1. (h) C_HP. (i) HyperPNN. (j) DRCNN. (k) KitNet. (l) DARN. (m) MDANet. (n) Ours_SR2. (o) EXP. (p) C_HMP. (q) CNTR. (r) LTCN. (s) HyperNet. (t) HMPNet. (u) Ours_SR3. The false color image is used for clear visualization (red: 35, green: 16, and blue: 7). The large red box area is obtained by enlarging the small red box area.

for the HIF. Therefore, these components of MGSAM can effectively improve the performance of HIF.

To investigate the effectiveness of the mutual learning strategy, a series of experiments are conducted, as shown in Table VII. When using the supervision of GTs and mutual learning at the beginning simultaneously (i.e., single stage), the results are lower compared with two stages, i.e., the first stage utilizes supervision of GTs and the second stage utilizes supervision of GTs and mutual learning. This is because incorporating the mutual learning strategy at the beginning might lead to more of a misguided knowledge transfer among the networks. There is an overall decrease in the evaluation metrics when the mutual learning strategy is removed. This is because the mutual learning strategy can prompt the sub-networks to transfer knowledge, resulting in better fusion performance. Meanwhile, the best fusion results are obtained when $\lambda$ is set to 0.1, which is due to the reason that too large $\lambda$ might lead to too much weakening of the supervision of GTs.

## V. CONCLUSION

In this article, UniNet is proposed for HIF, which can support different input conditions. The UniNet significantly improves the convenience and practical utility of HIF. Specifically, the overall framework consists of sub-network 1, sub-network 2, and the fusion network. In the training phase, the whole network is optimized. In the testing phase, three sub-networks can obtain HR HSIs under different input conditions, respectively. In addition to the supervision of GTs, a mutual learning strategy is proposed to transfer knowledge and experience, such as spatial and spectral features, among different sub-networks to obtain higher quality HSIs. In the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

MGSAM, we extract MS features employing convolution of different sizes first. Then, a GSA is designed to extract the spatial edge features, thus enriching the edge details of fused images. Experiments on four publicly available HSI datasets show that the proposed UniNet via this "all in one" manner achieves better results under three different input conditions compared to benchmarks (i.e., single HMF, HPF, and HMPF algorithms), respectively. In the future, we will explore combining image preprocessing, i.e., image registration, with HSI fusion, denoising, and other low-level tasks to improve HSI quality more conveniently.

## REFERENCES


[1] D. Liu, J. Li, and Q. Yuan, "A spectral grouping and attention-driven residual dense network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7711–7725, Sep. 2021.

[2] B. Yang, Y. Mao, L. Liu, X. Liu, Y. Ma, and J. Li, "From trained to untrained: A novel change detection framework using randomly initialized models with spatial–channel augmentation for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4402214.

[3] M. Wang, Q. Wang, D. Hong, S. K. Roy, and J. Chanussot, "Learning tensor low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 679–691, Jan. 2023.

[4] B. Tu, X. Yang, W. He, J. Li, and A. Plaza, "Hyperspectral anomaly detection using reconstruction fusion of quaternion frequency domain analysis," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 6, pp. 8358–8372, Jun. 2024.

[5] B. Yang, L. Qin, J. Liu, and X. Liu, "UTRNet: An unsupervised time-distance-guided convolutional recurrent network for change detection in irregularly collected images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410516.

[6] J. Chen, H. Liu, J. Hou, M. Yang, and M. Deng, "Improving building change detection in VHR remote sensing imagery by combining coarse location and co-segmentation," *ISPRS Int. J. Geo-Inf.*, vol. 7, no. 6, p. 213, Jun. 2018.

[7] H. Gao, Y. Yang, C. Li, L. Gao, and B. Zhang, "Multiscale residual network with mixed depthwise convolution for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3396–3408, Apr. 2021.

[8] H. Gao, H. Feng, Y. Zhang, S. Xu, and B. Zhang, "AMSSE-Net: Adaptive multiscale spatial–spectral enhancement network for classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5531317.

[9] H. Irmak, G. B. Akar, and S. E. Yuksel, "A MAP-based approach for hyperspectral imagery super-resolution," *IEEE Trans. Image Process.*, vol. 27, no. 6, pp. 2942–2951, Jun. 2018.

[10] X. Wang, Q. Hu, Y. Cheng, and J. Ma, "Hyperspectral image super-resolution meets deep learning: A survey and perspective," *IEEE/CAA J. Autom. Sinica*, vol. 10, no. 8, pp. 1668–1691, Aug. 2023.

[11] Y. Zheng et al., "Edge-conditioned feature transform network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5513315.

[12] J.-F. Hu, T.-Z. Huang, L.-J. Deng, H.-X. Dou, D. Hong, and G. Vivone, "Fusformer: A transformer-based fusion network for hyperspectral image super-resolution," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[13] L. He, J. Zhu, J. Li, A. Plaza, J. Chanussot, and B. Li, "HyperPNN: Hyperspectral pansharpening via spectrally predictive convolutional neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 8, pp. 3092–3100, Aug. 2019.

[14] P. Guan and E. Y. Lam, "Multistage dual-attention guided fusion network for hyperspectral pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5515214.

[15] X. Meng et al., "Integrated fusion for panchromatic, multispectral, hyperspectral remote sensing images with different swath widths," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[16] X. Tian, K. Li, W. Zhang, Z. Wang, and J. Ma, "Interpretable model driven deep network for hyperspectral, multispectral, and panchromatic image fusion," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, pp. 1–14, May 2023, doi: 10.1109/TNNLS.2023.3278928.

[17] M. T. Eismann and R. C. Hardie, "Application of the stochastic mixing model to hyperspectral resolution enhancement," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 9, pp. 1924–1933, Sep. 2004.

[18] N. Akhtar, F. Shafait, and A. Mian, "Bayesian sparse representation for hyperspectral image super resolution," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3631–3640.

[19] R. Dian, S. Li, and L. Fang, "Learning a low tensor-train rank representation for hyperspectral image super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2672–2683, Sep. 2019.

[20] W. He, Y. Chen, N. Yokoya, C. Li, and Q. Zhao, "Hyperspectral super-resolution via coupled tensor ring factorization," *Pattern Recognit.*, vol. 122, Feb. 2022, Art. no. 108280.

[21] L. Zhang, W. Wei, C. Bai, Y. Gao, and Y. Zhang, "Exploiting clustering manifold structure for hyperspectral imagery super-resolution," *IEEE Trans. Image Process.*, vol. 27, no. 12, pp. 5969–5982, Dec. 2018.

[22] J. Liu, Z. Wu, L. Xiao, J. Sun, and H. Yan, "A truncated matrix decomposition for hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 29, pp. 8028–8042, 2020.

[23] F. Palsson, J. R. Sveinsson, and M. O. Ulfarsson, "Multispectral and hyperspectral image fusion using a 3-D-convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 5, pp. 639–643, May 2017.

[24] X. Zhang, W. Huang, Q. Wang, and X. Li, "SSR-NET: Spatial–spectral reconstruction network for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5953–5965, Jul. 2021.

[25] S.-Q. Deng, L.-J. Deng, X. Wu, R. Ran, D. Hong, and G. Vivone, "PSRT: Pyramid shuffle- and-reshuffle transformer for multispectral and hyperspectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5503715.

[26] B. Pan, Q. Qu, X. Xu, and Z. Shi, "Structure–color preserving network for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5520512.

[27] G. Vivone, "Multispectral and hyperspectral image fusion in remote sensing: A survey," *Inf. Fusion*, vol. 89, pp. 405–417, Jan. 2023.

[28] L. Loncan et al., "Hyperspectral pansharpening: A review," *IEEE Geosci. Remote Sens. Mag. Replaces Newsletter*, vol. 3, no. 3, pp. 27–46, Sep. 2015.

[29] B. Aiazzi, S. Baronti, and M. Selva, "Improving component substitution pansharpening through multivariate regression of MS+Pan data," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 10, pp. 3230–3239, Oct. 2007.

[30] J. Qu, S. Hou, W. Dong, S. Xiao, Q. Du, and Y. Li, "A dual-branch detail extraction network for hyperspectral pansharpening," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518413.

[31] W. Dong, C. Zhou, F. Wu, J. Wu, G. Shi, and X. Li, "Model-guided deep hyperspectral image super-resolution," *IEEE Trans. Image Process.*, vol. 30, pp. 5754–5768, 2021.

[32] W. G. C. Bandara and V. M. Patel, "HyperTransformer: A textural and spectral feature fusion transformer for pansharpening," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2022, pp. 1767–1777.

[33] G. Vivone, A. Garzelli, Y. Xu, W. Liao, and J. Chanussot, "Panchromatic and hyperspectral image fusion: Outcome of the 2022 WHISPERS hyperspectral pansharpening challenge," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 166–179, 2023.

[34] N. Yokoya, T. Yairi, and A. Iwasaki, "Hyperspectral, multispectral, and panchromatic data fusion based on coupled non-negative matrix factorization," in *Proc. 3rd Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Jun. 2011, pp. 1–4.

[35] R. Arablouei, "Fusing multiple multiband images," *J. Imag.*, vol. 4, no. 10, p. 118, Oct. 2018.

[36] X. Tian, W. Zhang, Y. Chen, Z. Wang, and J. Ma, "HyperFusion: A computational approach for hyperspectral, multispectral, and panchromatic image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518216.

[37] K. Li, W. Zhang, D. Yu, and X. Tian, "HyperNet: A deep network for hyperspectral, multispectral, and panchromatic image fusion," *ISPRS J. Photogramm. Remote Sens.*, vol. 188, pp. 30–44, Jun. 2022.

[38] Y. Liang, S. Li, C. Yan, M. Li, and C. Jiang, "Explaining the black-box model: A survey of local interpretation methods for deep neural networks," *Neurocomputing*, vol. 419, pp. 168–182, Jan. 2021.

[39] G. Camps-Valls, D. Tuia, X. X. Zhu, and M. Reichstein, *Deep Learning for the Earth Sciences: A Comprehensive Approach to Remote Sensing, Climate Science and Geosciences*. Hoboken, NJ, USA: Wiley, 2021.


Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.

[40] Z. Zhou et al., "UNet++: Redesigning skip connections to exploit multiscale features in image segmentation," *IEEE Trans. Med. Imag.*, vol. 39, no. 6, pp. 1856–1867, Dec. 2019.

[41] J. Yang, C. Wu, B. Du, and L. Zhang, "Enhanced multiscale feature fusion network for HSI classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10328–10347, Dec. 2021.

[42] F. Zhou, X. Li, and Z. Li, "High-frequency details enhancing DenseNet for super-resolution," *Neurocomputing*, vol. 290, pp. 34–42, May 2018.

[43] H. Zhao, O. Gallo, I. Frosio, and J. Kautz, "Loss functions for image restoration with neural networks," *IEEE Trans. Comput. Imag.*, vol. 3, no. 1, pp. 47–57, Mar. 2017.

[44] Y. Zhang, T. Xiang, T. M. Hospedales, and H. Lu, "Deep mutual learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4320–4328.

[45] N. Yokoya and A. Iwasaki, "Airborne hyperspectral data over Chikusei," Dept. Space Appl. Lab., Univ. Tokyo, Tokyo, Japan, Tech. Rep. SAL-2016-05-27, 2016, vol. 5, p. 5.

[46] X. Huang and L. Zhang, "A comparative study of spatial approaches for urban mapping using hyperspectral ROSIS images over Pavia City, northern Italy," *Int. J. Remote Sens.*, vol. 30, no. 12, pp. 3205–3221, Jun. 2009.

[47] F. Yasuma, T. Mitsunaga, D. Iso, and S. K. Nayar, "Generalized assorted pixel camera: Postcapture control of resolution, dynamic range, and spectrum," *IEEE Trans. Image Process.*, vol. 19, no. 9, pp. 2241–2253, Sep. 2010.

[48] A. Chakrabarti and T. Zickler, "Statistics of real-world hyperspectral images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2011, pp. 193–200.

[49] L. Wald, T. Ranchin, and M. Mangolini, "Fusion of satellite images of different spatial resolutions: Assessing the quality of resulting images," *Photogramm. Eng. Remote Sens.*, vol. 63, no. 6, pp. 691–699, 1997.

[50] Q. Xie, M. Zhou, Q. Zhao, Z. Xu, and D. Meng, "MHF-Net: An interpretable deep network for multispectral and hyperspectral image fusion," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 3, pp. 1457–1473, Mar. 2020.

[51] Q. Wei, N. Dobigeon, and J.-Y. Tourneret, "Fast fusion of multi-band images based on solving a Sylvester equation," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4109–4121, Nov. 2015.

[52] S. Li, Y. Tian, C. Wang, H. Wu, and S. Zheng, "Hyperspectral image super-resolution network based on cross-scale nonlocal attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509615.

[53] Q. Yuan, Y. Wei, X. Meng, H. Shen, and L. Zhang, "A multiscale and multidepth convolutional neural network for remote sensing imagery pan-sharpening," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 978–989, Mar. 2018.

[54] Y. Zheng, J. Li, Y. Li, K. Cao, and K. Wang, "Deep residual learning for boosting the accuracy of hyperspectral pansharpening," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 8, pp. 1435–1439, Aug. 2020.

[55] W. G. C. Bandara, J. M. J. Valanarasu, and V. M. Patel, "Hyperspectral pansharpening based on improved deep image prior and residual reconstruction," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5520816.

[56] Y. Zheng, J. Li, Y. Li, J. Guo, X. Wu, and J. Chanussot, "Hyperspectral pansharpening using deep prior and dual attention residual network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 8059–8076, Nov. 2020.

[57] B. Aiazzi, L. Alparone, S. Baronti, and A. Garzelli, "Context-driven fusion of high spatial and spectral resolution images based on oversampled multiresolution analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 10, pp. 2300–2312, Oct. 2002.

[58] K. He, X. Zhang, S. Ren, and J. Sun, "Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1026–1034.

[59] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

[60] Q. Huynh-Thu and M. Ghanbari, "Scope of validity of PSNR in image/video quality assessment," *Electron. Lett.*, vol. 44, no. 13, pp. 800–801, 2008.

[61] W. Zhou, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, pp. 600–612, 2004.

[62] Y. Liu, J. Hu, X. Kang, J. Luo, and S. Fan, "Interactformer: Interactive transformer and CNN for hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531715.

[63] L. Wald, *Data Fusion: Definitions and Architectures: Fusion of Images of Different Spatial Resolutions*. Presses de l'École des Mines, 2002. [Online]. Available: https://www.google.com/books/edition/_/JOQ6elyzq-8C?hl=en&gbpv=0



**Yaoting Liu** received the B.S. degree in automation from the University of South China, Hengyang, China, in 2020, and the M.S. degree in control science and engineering from the School of Electrical and Information Engineering, Changsha University of Science and Technology, Changsha, China, in 2023. He is currently pursuing the Eng.D. degree with the College of Electrical and Information Engineering, Hunan University, Changsha, Hunan.

His research interests include remote sensing, data fusion, change detection, and deep learning.



**Xinwei Cheng** received the B.S. degree in automation from Nanchang University, Nanchang, China, in 2022. He is currently pursuing the M.S. degree in electronic information with Hunan University, Changsha, China.

His research interests include deep learning and remote sensing change detection.



**Hui Li** received the M.S. degree in environmental science from China University of Mining and Technology, Beijing, China, in 2013, and the Ph.D. degree in environmental science from Beijing Normal University, Beijing, in 2019.

She is currently an Associate Professor with the School of Environment, Beijing Normal University. Her research interests include the analysis of environmental systems.



**Shenglian Luo** received the B.S. degree in metal corrosion and protection from Hunan University, Changsha, China, in 1982, the M.S. degree in metal corrosion and protection from Beihang University, Beijing, China, in 1986, and the Ph.D. degree in environmental science and engineering from Central South University, Changsha, in 2006.

He is currently a Professor with the Key Laboratory of Jiangxi Province for Persistent Pollutants Control and Resources Recycle, Nanchang Hangkong University, Nanchang, China. His research interests include environmental risk assessment and environmental data analysis.



**Bin Yang** (Member, IEEE) received the B.S. degree in science and technology of remote sensing from Beihang University, Beijing, China, in 2012, and the Ph.D. degree in photogrammetry and remote sensing from Peking University, Beijing, in 2017.

He is currently an Associate Professor with the College of Electrical and Information Engineering and the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Hunan University, Changsha, China. His research interests include remote sensing and artificial intelligence.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:50:51 UTC from IEEE Xplore. Restrictions apply.